

UNIVERSITÀ DEGLI STUDI DI MILANO

FACOLTÀ DI SCIENZE E TECNOLOGIE

CLASSIFICATION OF HOLOMORPHIC FOLIATIONS ON ALGEBRAIC SURFACES

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A Pepe e Tita,
siete il regalo di compleanno più bello che abbia mai ricevuto.

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1 Introduction

The aim of this thesis is to present a complete classification of holomorphic foliations on algebraic surfaces. Such a classification is the result of the works of M. Brunella, M. McQuillan and L.G. Mendes, and the main references can be found in [1], [3], [4], [5], [13] [15], [16], [18].

Intuition. Informally speaking, a holomorphic foliation on a smooth complex surface is a geometric structure that decomposes the surface into a disjoint union of complex curves, called *leaves*. Around a generic point of the surface, a foliation can be described by a holomorphic vector field or a differential 1-form that defines a vector field. Integrating this vector field yields the leaves, which form a holomorphic family of immersed Riemann surfaces. Let us remark that there is no reason for the leaves to be algebraic curves: in general, they are not and there are very simple examples of this phenomenon, such as linear foliations on a torus. Singularities may occur at isolated points where the vector field generating the foliation vanishes, and their structure plays a vital role in the birational classification theory.

Motivation. The study of holomorphic foliations on complex algebraic surfaces plays a central role in several problems, in particular those related to the classification of algebraic varieties: indeed, in many ways foliations reflect the structure of the underlying surface, while also encoding new geometric and dynamical information through the behavior of their leaves. The classification of foliations from a birational perspective reveals rich analogies with the classification of algebraic surfaces. One of the major motivations for the birational classification of foliations stems from the theory of entire holomorphic curves, developed at first by F. Bogomolov. As highlighted in the work of Brunella and McQuillan (see [4]), entire curves $f : \mathbb{C} \to X$ on a surface X of general type become tangent to certain foliations on suitable birational models. Thus, understanding which foliations can possess an entire leaf on a surface of general type becomes relevant in tackling conjectures such as that of Green-Griffiths, which aims to prove that the union of all possible images of non-constant entire curves on a variety of general type X is not Zariski-dense. The strategy pioneered by McQuillan in the case of surfaces, and surveyed in [4], shows that entire curves can be lifted to coverings of the surface where they become tangent to a foliation, whose positivity properties ultimately constrain the image of such an entire curve. This insight provided an early pivotal motivation to classify foliations according to the positivity of their canonical bundles.

Main tools used for the classification. In this paragraph, we are going to give a more precise definition of a foliation and we are going to see all the tools needed for the classification. We define a *foliation* on a smooth complex surface X as the data of a rank 1 subsheaf of the tangent sheaf

$$0 \longrightarrow T_{\mathcal{F}} \to TX$$

such that the quotient $TX/T_{\mathcal{F}}$ is locally free outside of a finite set $Z \subset X$, called the *singular set of the foliation*. Thus, $T_{\mathcal{F}}$ is a line bundle.

Given a holomorphic foliation $\mathcal{F}$ on X with isolated singularities, we define its canonical bundle

$$K_{\mathcal{F}} = T^*_{\mathcal{F}}$$

as the dual of the tangent bundle. Sections of $K_{\mathcal{F}}$ can be interpreted as holomorphic 1-forms "along the leaves of $\mathcal{F}$". In analogy with Mori's program for algebraic surfaces (see [13]), it is natural to study $\mathcal{F}$ via the numerical and geometrical properties of $K_{\mathcal{F}}$.

More precisely, we introduce:

1. the *Kodaira dimension* $\mathrm{kod}(\mathcal{F})$, defined by the power of the asymptotic growth of $h^0(X, K_{\mathcal{F}}^{\otimes m})$ as a function of m;

2. the *numerical Kodaira dimension* $\nu(\mathcal{F})$, obtained via the Zariski decomposition of $K_{\mathcal{F}}$.

These quantities are defined respectively in Sections 3.4 and 3.5.

Both invariants take values in $\{-\infty, 0, 1, 2\}$ and satisfy

$$\mathrm{kod}(\mathcal{F}) \leq \nu(\mathcal{F});$$

we say that a foliation is *abundant* if $\mathrm{kod}(\mathcal{F}) = \nu(\mathcal{F})$. Geometrically speaking, this means that $K_{\mathcal{F}}^{\otimes n}$ has as many sections as its numerical properties lead us to expect. This is particularly relevant since in the classification of projective surfaces, abundance for K_X always holds. The abundance conjecture for projective varieties states that if X is a projective variety with mild singularities and such that K_X is nef, then it is also semiample, meaning that some positive power of it is base-point free. This has been proved in dimensions 2 and 3, while it is still a conjecture in higher dimension; it is also equivalent to $\nu(K_X) = \mathrm{kod}(K_X)$. In the case of foliations on surfaces abundance can sometimes fail: we will see later in this thesis an example of $\mathrm{kod}(\mathcal{F}) < \nu(\mathcal{F})$, and it can also happen that $\nu(\mathcal{F}) = \mathrm{kod}(\mathcal{F}) = 2$, but $K_{\mathcal{F}}$ is not semiample: see [15, Section IV.2]. In this thesis, when we talk about abundance for a foliation $\mathcal{F}$, we will mean that $\mathrm{kod}(\mathcal{F}) = \nu(\mathcal{F})$.

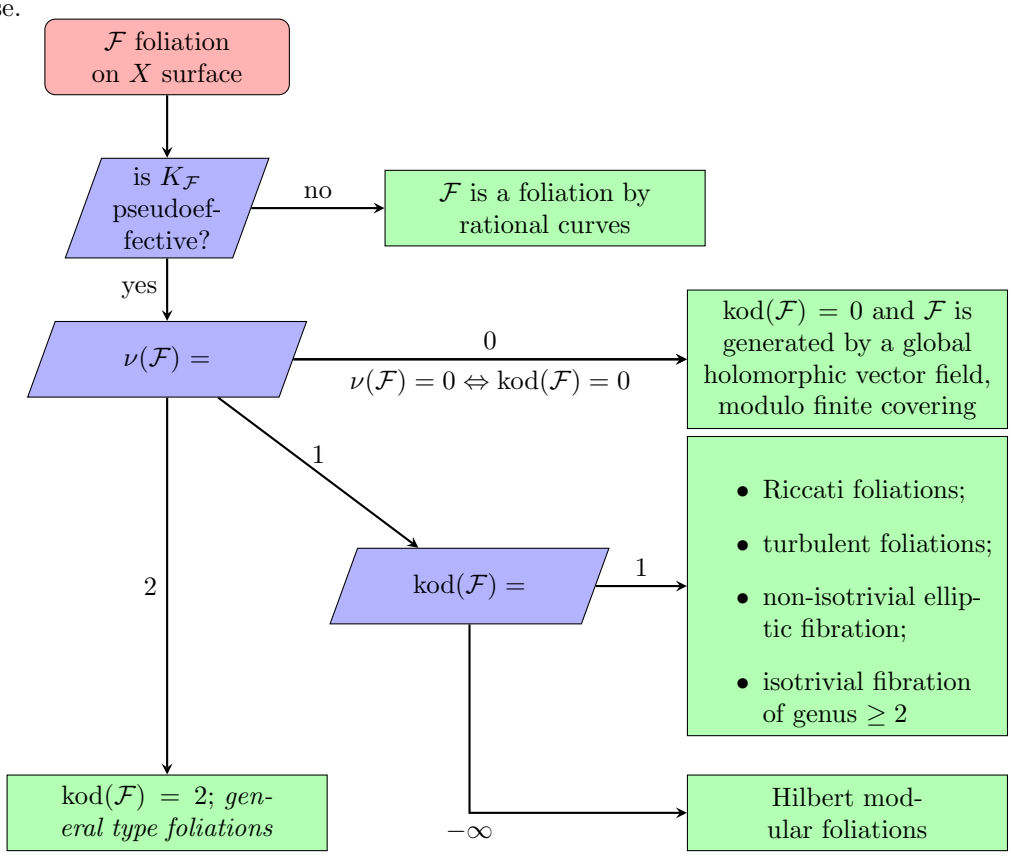
To ensure that these quantities are birational invariants, we must restrict the singularities of $\mathcal{F}$ to certain classes. We follow two approaches:

1. work with *reduced singularities*, relying on Seidenberg's theorem (Theorem 2.1.1) to ensure that every foliation is birationally equivalent to one with only reduced singularities; even if we blow up infinitely many times in a singular point of a foliation, it is not always possible to obtain smoothness, and the mildest singularities we can obtain are reduced ones.

2. extend our framework to *canonical singularities*, since these are the largest class of singularities such that the sections of multiples of the canonical bundle $H^0(X, K_{\mathcal{F}}^{\otimes n})$ do not change. The use of canonical foliated singularities will force us to consider also quotient singularities on surfaces of Hirzebruch-Jung type.

Firstly, we develop the theory under the the reduced-singularity hypothesis, then we switch to the canonical setting to deal with non-abundant foliations.

Summary of the classification We are now going to present the classification of foliations up to Kodaira and numerical Kodaira dimensions in a diagram, and then we are going to briefly explain the strategies used to obtain it case by case.



Foliations by rational curves In Section 2, we see a theorem due to Miyaoka ([3], Theorem 7.1) which characterizes foliations by rational curves from the point of view of the numerical dimension of the canonical bundle: indeed, these are the only foliations with non-pseudoeffective canonical bundle. Since the Zariski decomposition can only be applied when $K_{\mathcal{F}}$ is pseudoeffective, we set the numerical Kodaira dimension of such foliations to be $-\infty$. This forces the Kodaira dimension to be negative as well.

Foliations with non-negative Kodaira dimension. The classification of foliations with non-negative Kodaira dimension on an algebraic surface X fol-

lows four steps:

1. We classify all the foliations generated by a vector field with isolated zeroes that is globally holomorphic on X.

2. We see in 3.4 that a foliation $\mathcal{F}$ such that $\nu(\mathcal{F}) = 0$ is, up to a ramified covering and birational maps, generated by a global holomorphic vector field with isolated zeroes (for a complete proof, see [3], Theorem 8.2).

3. We see in Section 3.5.2 two particular cases of abundance: $\nu(\mathcal{F}) = 0$ if and only if $\mathrm{kod}(\mathcal{F}) = 0$ (see [3], Theorem 9.2 for a complete proof), and $\nu(\mathcal{F}) = 2$ if and only if $\mathrm{kod}(\mathcal{F}) = 2$, which follows from Riemann-Roch theorem.

4. From the previous points, clearly if $\mathrm{kod}(\mathcal{F}) = 1$ then also $\nu(\mathcal{F}) = 1$ and by Theorem 3.5.1 we have that $\mathcal{F}$ belongs to one of the following classes (we recall that a fibration is *isotrivial* if it becomes isomorphic to a product after taking a finite unramified cover):

   (a) Riccati foliations (see Section 2.5.2);

   (b) Turbulent foliations (see Section 2.5.4);

   (c) Non-isotrivial elliptic fibrations;

   (d) Isotrivial fibrations of genus ≥ 2.

Non-abundant foliations. The most interesting part of the classification is obtained in the non-abundant case, which from the previous points can only occur if $\nu(\mathcal{F}) = 1$ and $\mathrm{kod}(\mathcal{F}) = -\infty$: indeed, we define a Hilbert modular surface X as the (smooth) compactification of the resolution of the quotient of $\mathbb{D}\times\mathbb{D}$ by a subgroup $\Gamma \subset Aut(\mathbb{D}\times\mathbb{D})$; on X, we have two foliations $\mathcal{F}$ and $\mathcal{G}$ which naturally arise from the two projections on $\mathbb{D}\times\mathbb{D}$. It is not difficult to prove that $\mathcal{F}$ and $\mathcal{G}$, which we denominate *Hilbert modular foliations* are non-abundant, so this class is non-empty; however, it is more difficult to prove that all non-abundant foliations are Hilbert modular. Given a non-abundant foliation $\mathcal{F}$, the strategy to prove that it is Hilbert modular is to find its "companion foliation" $\mathcal{G}$; in Brunella's book [3], the program proposed consists in defining $\mathcal{G}$ as the kernel of a closed positive current Ω, such that $\Omega \wedge \Omega \equiv 0$. Nonetheless, there are some obstructions to this method: in particular, we do not know whether Ω is sufficiently regular to define the wedge product (see [7], Corollary 2.19), and even if it is there is no guarantee that $\mathcal{G}$ is a holomorphic foliation. However, in [3] it is proved that, if we can define $\mathcal{G}$ as we described in this paragraph, and it is a holomorphic foliation, then $\mathcal{F}$ and $\mathcal{G}$ are Hilbert modular foliations.

For a full proof of this fact, we follow Brunella's work in [5]: the strategy of this article is to define the Poincaré metric on each leaf of the non-abundant foliation $\mathcal{F}$, and to prove that the variation of the metric on the leaves has positive curvature both in the leaf-wise direction and in the transverse one, provided that the canonical bundle $K_{\mathcal{F}}$ is nef. It is precisely in this context that

we need to enlarge the class of singularities of the foliated surface to canonical ones: indeed, to obtain a nef $K_{\mathcal{F}}$, we might need to contract some curves and thus we get a new surface that presents Hirzebruch-Jung singularities, which are canonical. The main result of this paper ([5], Main Theorem), proves that the Poincaré metric defined on the canonical bundle $K_{\mathcal{F}}$ has positive curvature Ω; this allows us to apply the program described in this paragraph and we report in Section 4 the results of [5], Section 4, where it is proved that, with the canonical Poincaré metric, we can define a holomorphic foliation $\mathcal{G}$ as the kernel of Ω which is precisely the companion foliation of $\mathcal{F}$, making both $\mathcal{F}$ and $\mathcal{G}$ Hilbert modular foliations. In this way, non-abundant foliations are completely classified as Hilbert modular foliations.

Overview of the structure. 2 introduces all the tools necessary for the classification of foliations. We begin by defining foliations and their singularities, and we present the crucial adjunction formulae. After discussing some basic examples, we introduce closed positive currents and Hirzebruch–Jung singularities. This is followed by a discussion of several key examples of foliations that play an essential role in the classification. The chapter concludes with an analysis of foliations generated by global holomorphic vector fields.

3 is devoted to the classification of foliations, following the framework established in [3]. We start by recalling the minimal model program for algebraic surfaces, which serves as a foundation for the analogous program in the context of foliations. We then define minimal models for foliations and classify all foliations that do not admit a minimal model.

In 3.3, we present Miyaoka's rationality criterion, which provides a numerical characterization of rational fibrations as precisely those foliations whose canonical bundle is not pseudo-effective. 3.4 introduces the numerical Kodaira dimension $\nu(\mathcal{F})$ and includes a classification of foliations with vanishing numerical Kodaira dimension. We also define canonical singularities in this context.

At the beginning of 3.5, we define the Kodaira dimension of a foliation $\mathrm{kod}(\mathcal{F})$ and prove the inequality $\mathrm{kod}(\mathcal{F}) \leq \nu(\mathcal{F})$. The remainder of the section is devoted to the classification of foliations with Kodaira dimension 0 and 1. In Section 3.6, we study Hilbert modular foliations, showing that they are non-abundant.

4 is devoted to the study of non-abundant foliations; in 4.1, we study Hilbert modular foliations and we show that they are non-abundant; the remaining part of the Section is dedicated to proving that non abundant foliations are all Hilbert modular. Following the strategy of [5], we introduce a singular Hermitian metric on the canonical bundle $K_{\mathcal{F}}$, obtained by equipping each leaf with the Poincaré metric. This construction yields the so-called canonical Poincaré metric Ω, and we expose the main result of [5], i.e. that Ω has positive curvature. This allows us to apply the program discussed before and prove that $\mathcal{F}$ is a Hilbert modular foliation.

2 Preliminaries

2.1 Definition and first properties

In this section, our aim is to give some context about foliations. We will see in details the definition of a foliation and we will study the singularities that a foliation can present: indeed, in the next section we shall find some birational invariants for foliations, provided that the singularities are of a certain kind, namely reduced singularities. We will see what happens to the singularities in case of a blow-up, and then we will move on to the adjunction formulae, which will help us to deduce useful properties of specific foliations throughout the whole thesis. We will see some examples to make the concepts clearer. Finally, before moving on to the next section, we will see examples of very important foliations that will play a decisive role in the classification. Our main reference for this chapter will be [3], unless indicated otherwise.

2.1.1 Intrinsic definition

Definition 2.1.1. *Let X be a smooth complex surface. A foliation $\mathcal{F}$ on X is given by a rank one subsheaf of the tangent sheaf*

$$0 \longrightarrow T_{\mathcal{F}} \longrightarrow TX,$$

such that the quotient sheaf $TX/T_{\mathcal{F}}$ is locally free outside of a finite set Z. From this definition, it immediately follows that $T_{\mathcal{F}}$ is a line bundle, being X a smooth surface.

We call $T_{\mathcal{F}}$ the *tangent bundle* of the foliation $\mathcal{F}$; we can also consider its dual $T^*_{\mathcal{F}} =: K_{\mathcal{F}}$, and we call this the *canonical bundle of the foliation*.

Considering the quotient sheaf $TX/T_{\mathcal{F}}$, we can fit the tangent bundle into the short exact sequence

$$0 \longrightarrow T_{\mathcal{F}} \longrightarrow TX \longrightarrow \mathcal{I}_Z \cdot N_{\mathcal{F}} \longrightarrow 0,$$

where $\mathcal{I}_Z \subset \mathcal{O}_X$ is the ideal sheaf supported on Z and $N_{\mathcal{F}}$ is a line bundle, called the *normal bundle* of the foliation $\mathcal{F}$.

The idea behind the classification of holomorphic foliations is to analyze the properties of these line bundles and deduce, from them, properties of the foliation $\mathcal{F}$.

We call Z the *singular set* of the foliation $\mathcal{F}$ and denote it with $\mathrm{Sing}(\mathcal{F})$. This set will be of utmost importance in our thesis, since we are going to define some quantities that turn out to be birational invariants only if the singularities are of a certain type.

Thus, outside of the singular set $\mathrm{Sing}(\mathcal{F})$, a foliation associates a tangent direction to each point. Thanks to Frobenius Theorem, we can prove that $T_{\mathcal{F}}$ is integrable, i.e. through each point p of X there exists an integral manifold L_p of $\mathcal{F}$, called *leaf of $\mathcal{F}$ through p*. Indeed, a foliation is an involutive distribution: since $T_{\mathcal{F}}$ is a subsheaf of TX, it is a tangent distribution, and since it is a line

bundle it is closed under Lie bracket and by definition involutive. To prove this, observe that $T_{\mathcal{F}_p}$ is a linear subspace of T_pX of dimension 1, and thus presents a basis $\{v_p\}$; then, every $w_p, u_p \in T_{\mathcal{F}_p}$ can be written as $w_p = fv_p$, $u_p = gv_p$ for some $f, g \in \mathcal{O}_{X,p}$. If we compute the Lie bracket we obtain

$$[w_p, u_p] = [fv_p, gv_p] = fv_p(g)v_p - gv_p(f)v_p = (fv_p(g) - gv_p(f))v_p,$$

which is again in $T_{\mathcal{F}_p}$. Now we can use Frobenius theorem (see [14], chapter 14), which states that every involutive distribution is integrable, to prove that $T_{\mathcal{F}}$ is integrable. Keep in mind that the leaves of the foliation are holomorphic curves, but they might fail to be algebraic; we will see an example of this later on.

2.1.2 Definition on open subsets

We can give an equivalent definition of foliation which is easier to work with locally. Let X be a smooth complex surface and let $\{U_j\}_{j\in I}$ be an open covering of X. On each U_j, we take v_j as a holomorphic vector field with isolated zeroes and we require that on each $U_i \cap U_j$, $v_i = g_{ij}v_j$, where $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$. This means that the local integral curves of v_i and v_j glue together, up to reparametrization, giving the leaves of $\mathcal{F}$. We define an equivalence relation: $\{U_j, v_j\}_{j\in I} \sim \{U'_j, v'_j\}_{j\in I'}$ if and only if on $U_j \cap U'_i$, $v'_i = f_{ij}v_j$, where $f_{ij} \in \mathcal{O}_X^*(U'_i \cap U_j)$. We define a foliation $\mathcal{F}$ on X as an equivalence class of collections $\{U_j, v_j\}_{j\in I}$ modulo $\sim$.

We can define the singular set $Sing(\mathcal{F})$ of the foliation $\mathcal{F}$ on a smooth complex surface X locally as $Sing(\mathcal{F}) \cap U_j = \{\text{zeroes of } v_j\}$ for all $j \in I$.

We can re-obtain the tangent bundle $T_{\mathcal{F}}$ of the foliation by noticing that the functions $g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$ form a multiplicative cocycle, thus we can define a line bundle in X, which is precisely the canonical bundle of $\mathcal{F}$, which is also intrinsically defined by $\mathcal{F}$, since if we take $\{U'_j, v'_j\}_{j\in I'}$, we obtain a cocycle $\{g'_{ij}\}$ cohomologous to $\{g_{ij}\}$. We can also consider the dual $T_{\mathcal{F}}$, the tangent bundle of $\mathcal{F}$. The relations $v_i = g_{ij}v_j$ can be thought as the defining relations of a global holomorphic section $s \in T^*_{\mathcal{F}} \otimes TX$, that has isolated zeroes but is defined only up to a non-vanishing global holomorphic function f. We can see the section s as a morphism $T_{\mathcal{F}} \to TX$ which is injective over the regular points of $\mathcal{F}$ and trivial over $Sing(\mathcal{F})$. Hence, it fits into an exact sequence

$$0 \longrightarrow T_{\mathcal{F}} \longrightarrow TX \longrightarrow \mathcal{I}_Z \cdot N_{\mathcal{F}} \longrightarrow 0, \tag{1}$$

and we obtain precisely the short exact sequence we saw earlier in the chapter.

We can also use 1-forms instead of vector fields and define a foliation as a collection of 1-forms $\omega_j \in \Omega^1_X(U_j)$ with isolated zeroes and such that $\omega_i = f_{ij}\omega_j$ on $U_i \cap U_j$, $f_{ij} \in \mathcal{O}_X^*(U_i \cap U_j)$. The cocycle f_{ij} defines a line bundle on X, called normal bundle of $\mathcal{F}$ and identifiable with the line bundle $N_{\mathcal{F}}$ in the sequence above. The foliation now gives a global holomorphic section of $N_{\mathcal{F}} \otimes T^*X$, with discrete zero set and modulo multiplication by $\mathcal{O}_X^*(X)$, and to an exact

sequence

$$0 \longrightarrow N_{\mathcal{F}}^* \longrightarrow T^*X \longrightarrow \mathcal{I}_Z \cdot T_{\mathcal{F}}^* \longrightarrow 0. \tag{2}$$

These line bundles are related to each other by the canonical bundle of X:

$$K_X = T_{\mathcal{F}}^* \otimes N_{\mathcal{F}}^*$$

In fact, we know that, whenever we have an exact sequence of vector bundles

$$0 \longrightarrow E \longrightarrow F \longrightarrow F/E \longrightarrow 0,$$

the formula

$$det(F) = det(E) \otimes det(F/E)$$

holds, and we can apply it to (2) to obtain

$$det(T^*X) = det(N_{\mathcal{F}}^*) \otimes det(\mathcal{I}_Z \cdot T_{\mathcal{F}}^*);$$

the reader should keep in mind that $K_X = det(T^*X)$ and, since Z is a finite discrete set, $det(\mathcal{I}_Z \cdot T_{\mathcal{F}}^*) = det(T_{\mathcal{F}}^*)$; moreover, since $N_{\mathcal{F}}^*$ and $T_{\mathcal{F}}^*$ are line bundles, their determinant coincides with them.

2.1.3 Reduced singularities

At the beginning of this section, we mentioned our goal to define certain birational invariants for a foliation $\mathcal{F}$, i.e. the Kodaira dimension $\text{kod}(\mathcal{F})$ and the numerical Kodaira dimension $\nu(\mathcal{F})$. As stated before, for those dimensions to be birational invariants, we need to work with special kinds of singularities: in particular, we will define *reduced singularities*.

Let $\mathcal{F}$ be a foliation on a complex surface X, locally defined by a holomorphic vector field v or, equivalently, by a holomorphic 1-form ω, and let $p \in X$ be an isolated singular point of $\mathcal{F}$.

Definition 2.1.2. *An irreducible curve C on a neighborhood of p is said to be invariant by $\mathcal{F}$ (or $\mathcal{F}$-invariant) if v is tangent to C or, equivalently, if the pullback of ω to C is identically zero.*

Definition 2.1.3. *A separatrix of $\mathcal{F}$ at p is a holomorphic, possibly singular irreducible curve C on a neighborhood of p which passes through p and is invariant by $\mathcal{F}$.*

Intuitively, a separatrix through p is a leaf germ invariant under the foliation. Since a foliation is an integrable distribution, at least one separatrix passes through each point of the surface X. In some cases, we can have infinitely many separatrices through a singular point, in which case the singularity is called *dicritical*.

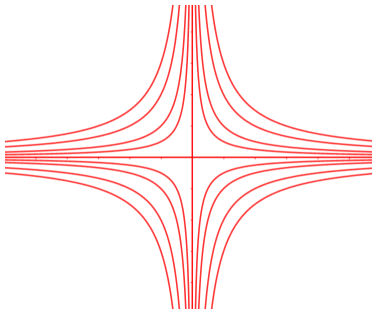
If p is a singular point of v, then we can consider the linear part $(Dv)(p)$ and its eigenvalues λ_1, λ_2 of v at p are well defined since $v(p) = 0$.

Definition 2.1.4. *The singularity p is called a reduced singularity if at least one of these eigenvalues, say λ_2, is not zero and the quotient $\lambda = \frac{\lambda_1}{\lambda_2}$ is not a positive rational number.*

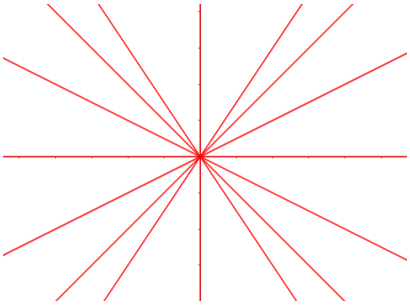
Remark 2.1.1. The quotient λ is unchanged by the multiplication of v by a non-vanishing holomorphic function, so we may speak of reduced singularity of a foliation.

If λ_1 is also not zero, then the reduced singularity is called *non-degenerate*. Otherwise, p is a *saddle-node*.

Example 2.1.1. Let $\mathcal{F}$ be the foliation defined on $\mathbb{C}^2$ by the vector field $v = z\frac{\partial}{\partial z} - w\frac{\partial}{\partial w}$. The associated 1-form is $\omega = wdz + zdw$. The foliation has a singularity in $p = (0,0)$ and $(Dv)(p) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, therefore the eigenvalues are $\lambda_1 = 1,\ \lambda_2 = -1$ and so $\lambda = -1$, making p a reduced singularity.



Example 2.1.2. Let $\mathcal{F}$ be the foliation defined on $\mathbb{C}^2$ by the vector field $v = z\frac{\partial}{\partial z} + w\frac{\partial}{\partial w}$. The associated 1-form is $\omega = wdz - zdw$. The foliation has a singularity in $p = (0,0)$ and $(Dv)(p) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, therefore the eigenvalues are $\lambda_1 = \lambda_2 = 1$ and so $\lambda = 1$, making p a non-reduced singularity. In fact, the singularity is dicritical: the leaves of this foliation are given by all the lines through p, which are of course invariant by $\mathcal{F}$ and are holomorphic irreducible curves.



This particular foliation is called *radial foliation*.

2.1.4 Blow-ups and singularities

In this section, we investigate the behavior of the foliation under blow-ups of the surface X at singular points of the foliation $\mathcal{F}$. In order to do so, we will consider the local description. Let $\mathcal{F}$ be a foliation on a surface X, locally generated by a vector field v or a 1-form ω with isolated zeroes. Let p be a singular point of $\mathcal{F}$, and let $\pi : \widetilde{X} \to X$ be the blow-up of X at p, with exceptional divisor $E = \pi^{-1}(p) \simeq \mathbb{C}P^1$. We want to define a foliation $\widetilde{\mathcal{F}} = \pi^*(\mathcal{F})$: the 1-form $\widetilde{\omega} = \pi^*(\omega)$ is still holomorphic, but its zero locus is not discrete anymore, since it contains E. We denote with $l(p) \in \mathbb{N}^+$ the vanishing order of $\widetilde{\omega}$ on E and with $a(p)$ the vanishing order of ω in p. Near each point of E, we can divide by the $l(p)$-power of a local equation of E, thus obtaining a 1-form with isolated zeroes. On the intersection of two neighborhoods, the corresponding 1-forms differ by multiplication by a nowhere vanishing holomorphic function, therefore we have a global foliation $\widetilde{\mathcal{F}}$ on $\widetilde{X}$.

Let us observe that if E is not $\widetilde{\mathcal{F}}$-invariant, then we have $l(p) = a(p) + 1$: indeed, the non-invariant case corresponds to a further degeneration of $\widetilde{\omega}$. If E is $\widetilde{\mathcal{F}}$-invariant we have $l(p) = a(p)$.

Example 2.1.3. We consider the foliation $\mathcal{F} \leftrightarrow v = z\frac{\partial}{\partial z} - w\frac{\partial}{\partial w} \leftrightarrow \omega = zdw + wdz$ on $\mathbb{C}^2$ and we blow-up in $p = (0, 0)$, obtaining the surface $\pi : X \to \mathbb{C}P^2, \quad E \mapsto p$. By the local description of the blow-up, we know that in a neighborhood of p we can write X as

$$X = \{((z, w), [S : T]) \in \mathbb{C}^2 \times \mathbb{C}P^1 \mid zT = wS\}.$$

We have that $X = \widetilde{U}_0 \cup \widetilde{U}_1$, where

$$\widetilde{U}_0 = \{((z, w), [S : T]) \in X \mid S \neq 0\}$$

$$\widetilde{U}_1 = \{((z, w), [S : T]) \in X \mid T \neq 0\}$$

are the local charts of the atlas covering X. Suppose that we are in the local chart $\widetilde{U}_0$; then, from the relations $S \neq 0$ and $zT = wS$ we obtain $w = \frac{T}{S}z =: tz$, therefore the map $\pi\big|_{\widetilde{U}_0} =: \pi_0$ can be locally described as the map

$$\pi_0 : (z, t) \longmapsto (z, tz).$$

In the other chart $\widetilde{U}_1$, we can describe the blow-up as

$$\pi_1 : (s, w) \longmapsto (ws, w),$$

where $s := \frac{S}{T}$.

If we now pull back the holomorphic 1-form ω, we obtain

$$\pi_0^*\omega = zd(tz) + tzdz = 2tzdz + z^2dt = z(2tdz + zdt);$$

we can repeat the computations in the other chart $\widetilde{U}_1$ and we get

$$\pi_1^*\omega = 2swdw + w^2ds = w(2sdw + wds).$$

We define the new foliation $\widetilde{\mathcal{F}}$ as the kernel of the holomorphic 1-forms $\{\frac{\pi_0^*\omega}{z}, \frac{\pi_1^*\omega}{w}\}$. If we take $\{z=0\}$ as a local equation for E, we immediately see that $l(p) = a(p) = 1$, therefore E is $\widetilde{\mathcal{F}}$-invariant, thus a leaf of the foliation $\widetilde{\mathcal{F}}$.

Example 2.1.4. We consider the radial foliation $\mathcal{F} \leftrightarrow v = z\frac{\partial}{\partial z} + w\frac{\partial}{\partial w} \leftrightarrow \omega = zdw - wdz$ on $\mathbb{C}^2$ and we blow-up again in $p = (0,0)$, obtaining the surface $\pi : X \to \mathbb{C}P^2, \quad E \mapsto p$. Using the same local description of the blow-up as before, we obtain

$$\pi_0^*\omega = zd(tz) - tzdz = z^2dt;$$

$$\pi_1^*\omega = wd(ws) - wsdw = w^2ds.$$

We define the foliation on the blown-up surface as $\widetilde{\mathcal{F}} = ker\{\frac{\pi_0^*\omega}{z^2}, \frac{\pi_1^*\omega}{w^2}\}$ and taking again $E = \{z=0\}$ locally, we see that $l(p) = 2 = a(p) + 1$, so E is not $\widetilde{\mathcal{F}}$-invariant. We can see, however, that the foliation $\widetilde{\mathcal{F}}$ is not singular anymore.

As we stated before, reduced singularities are fundamental in our classification. Fortunately, a result due to Seidenberg proves that we can always find a resolution of the foliated surface $(X, \mathcal{F})$ where all the singularities are reduced:

Theorem 2.1.1 (Seidenberg)**.** *Given any singular point $p \in X$ of $\mathcal{F}$, there exists a sequence of blow-ups $\hat{\pi} : \hat{X} \to X$ over p such that the foliation $\hat{\mathcal{F}} = \hat{\pi}^*(\mathcal{F})$ has only reduced singularities over $\hat{\pi}^{-1}(p)$.*

The proof of this theorem can be found in [18, Theorem 12].

Let us see what happens when we blow up a reduced singularity p. If p is non-degenerate with eigenvalue $\lambda \neq 0$, then the exceptional divisor E is $\widetilde{\mathcal{F}}$-invariant and contains two singularities p_1, p_2, both reduced and non-degenerate. In fact, if the exceptional divisor were not $\widetilde{\mathcal{F}}$-invariant, then it would intersect with infinitely many leaves of $\widetilde{\mathcal{F}}$ and thus when we contract E again we would have infinitely many separatrices through p, making p a dicritical and therefore non-reduced singularity. The strict transforms of two separatrices through p meet E in two distinct points, which are the singularities of $\widetilde{\mathcal{F}}$.

If p is a saddle-node, then the exceptional divisor E is again $\widetilde{\mathcal{F}}$-invariant and contains two singularities, a saddle-node p_1 and a reduced non-degenerate singularity of eigenvalue -1.

As a consequence to this analysis and Seidenberg Theorem, we obtain the next proposition:

Proposition 2.1.1. *[3, Proposition 1.1] A singularity p of a foliation $\mathcal{F}$ is dicritical if and only if there exists a sequence of blow-ups $\hat{\pi} : \hat{U} \to U$ over p and an irreducible component E_0 of $\hat{\pi}^{-1}(p) = \hat{E}$ which is not invariant by $\hat{\mathcal{F}} = \hat{\pi}^*(\mathcal{F})$.*

2.2 Adjunction formulae

In order to understand the geometric meaning of $T_{\mathcal{F}}$ and $N_{\mathcal{F}}$, we give some formulae concerning the computation of their degrees on compact curves.

Let $\mathcal{F}$ be a foliation on the surface X, not necessarily compact, and let $C \subset X$ be a compact curve, possibly singular. If each irreducible component C is not invariant by $\mathcal{F}$, and p is any point of C, we define an index $tang(\mathcal{F}, C, p)$ as follows: let $\{f = 0\}$ be a local reduced equation of C around p, let v be a holomorphic vector field with isolated zeroes generating $\mathcal{F}$ around p, and set

$$tang(\mathcal{F}, C, p) = dim_{\mathbb{C}} \frac{\mathcal{O}_{X,p}}{< f, v(f) >},$$

where $\mathcal{O}_p$ is the local algebra of X at p, $v(f)$ is the Lie derivative of f along v, and $< f, v(f) >$ is the ideal generated by $f, v(f)$ in $\mathcal{O}_{X,p}$. Since C is not $\mathcal{F}$-invariant, $v(f)$ is not identically zero on C, ensuring that the ideal $< f, v(f) >$ has finite codimension in $\mathcal{O}_{X,p}$ and $tang(\mathcal{F}, C, p) < \infty$. In fact, $tang(\mathcal{F}, C, p) = 0$ for all p where $\mathcal{F}$ is transverse on C. Hence, we can define

$$tang(\mathcal{F}, C) = \sum_{p \in C} tang(\mathcal{F}, C, p).$$

The adjunction formula allows us to define the virtual Euler characteristic $\chi(C)$ of C as follows:

$$\chi(C) = -K_X \cdot C - C \cdot C,$$

which coincides with the topological Euler characteristic if C is smooth, but is strictly smaller if C has singularities.

Proposition 2.2.1. *[3, Proposition 2.2] Let $\mathcal{F}$ be a foliation on a complex surface X and let C be a compact curve, each component of which is not invariant by $\mathcal{F}$. Then,*

$$N_{\mathcal{F}} \cdot C = \chi(C) + tang(\mathcal{F}, C);$$
$$T_{\mathcal{F}} \cdot C = C \cdot C - tang(\mathcal{F}, C).$$

Let us observe that these two formulae are equivalent, since $\chi(C) + C \cdot C = -K_X \cdot C$ and $K_X^* = T_{\mathcal{F}} \otimes N_{\mathcal{F}}$. If C is everywhere transverse to $\mathcal{F}$, then we obtain $T_{\mathcal{F}} \cdot C = C \cdot C$ and $N_{\mathcal{F}} \cdot C = \chi(C)$. Another straightforward consequence of the proposition is that if C is not $\mathcal{F}$-invariant, then $T_{\mathcal{F}}^* \cdot C + C \cdot C \geq 0$, with equality if and only if C is transverse to $\mathcal{F}$.

In case C is invariant by $\mathcal{F}$, we can introduce another index $Z(\mathcal{F}, C, p)$ for any $p \in C$: as before, let $\{f = 0\}$ be a local equation of C around p and let ω be a holomorphic 1-form generating $\mathcal{F}$ around p. Since C is $\mathcal{F}$-invariant, we can factorize ω around p as

$$g\omega = hdf + f\eta,$$

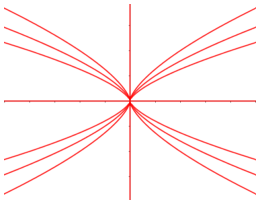
where η is a holomorphic 1-form, g and h are holomorphic functions and h, f and g, f are relatively prime. The function $\frac{h}{g}\big|_C$ is a meromorphic function, not everywhere equal to 0 or ∞, which does not depend on the choice of g, h, η since $\frac{h}{g}\big|_C$ is the residue of $\frac{\omega}{f}$ along C but outside of p. $\frac{h}{g}\big|_C$ depends on the choice of

ω and f, but if we change those we obtain $\frac{h'}{g'}\big|_C = l\frac{h}{g}\big|_C$, where l is a nowhere vanishing holomorphic function. Therefore, we can set

$$Z(\mathcal{F}, C, p) = \text{vanishing order of } \frac{h}{g}\Big|_C \text{ at } p = \sum_i \{\text{vanishing order of } \frac{h}{g}\Big|_{C_i} \text{ at } p\},$$

where $C_i \subset C$ are the irreducible components of C at p. It can happen that $\frac{h}{g}\big|_C$ has a pole in p and so $Z(\mathcal{F}, C, p) < 0$:

Example 2.2.1. Let $\mathcal{F}$ be the foliation defined around $p = (0, 0)$ as the kernel of the holomorphic 1-form $\omega = 2wdz - 3zdw$. The associated vector field is $v = 2w\frac{\partial}{\partial w} + 3z\frac{\partial}{\partial z}$, so $Dv(p) = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, so $\lambda = \frac{3}{2} \in \mathbb{Q}^+$ and p is not a reduced singularity. The $\mathcal{F}$-invariant curves are given by the equations $w^3 - kz^2 = 0$ for $k \in \mathbb{C}$, so the curve $C = \{w^3 - z^2 = 0\}$ is invariant for $\mathcal{F}$.



Let us calculate $Z(\mathcal{F}, C, p)$: keeping in mind the factorization of ω around p, we can choose η as $2dz + 3dw$ and $\frac{h}{g}\big|_C = -\frac{w+z}{w^2+z}\big|_C$; the vanishing order of $\frac{h}{g}\big|_C$ around p is therefore -1, since $\frac{h}{g}\big|_C$ has a first order pole at p. Thus, $Z(\mathcal{F}, C, p) = -1$.

Proposition 2.2.2. *If p is a non-dicritical singularity of $\mathcal{F}$, then $Z(\mathcal{F}, C, p) \geq 0$.*

Assuming now that C is compact, we can set

$$Z(\mathcal{F}, C) = \sum_{p \in C} Z(\mathcal{F}, C, p).$$

Proposition 2.2.3. *[3, Proposition 2.3] Let $\mathcal{F}$ be a foliation on a complex surface X and let $C \subset X$ be a compact curve, each component of which is invariant by $\mathcal{F}$. Then,*

$$N_{\mathcal{F}} \cdot C = C \cdot C + Z(\mathcal{F}, C);$$

$$T_{\mathcal{F}} \cdot C = \chi(C) - Z(\mathcal{F}, C).$$

We are interested in knowing what happens if we apply the previous propositions to the curve E. If E is $\widetilde{\mathcal{F}}$-invariant, then from Proposition 2.2.4 and $E \cdot E = -1$, we get

$$Z(\widetilde{\mathcal{F}}, E) = N_{\widetilde{\mathcal{F}}} \cdot E - E \cdot E = l(p) + 1 = a(p) + 1.$$

If E is not $\widetilde{\mathcal{F}}$-invariant, then from Proposition 2.2.2 we have

$$tang(\widetilde{\mathcal{F}}, E) = T^*_{\widetilde{\mathcal{F}}} \cdot E + E \cdot E = l(p) - 2 = a(p) - 1.$$

Example 2.2.2. Let $\mathcal{F}$ be the foliation given by $v = z\frac{\partial}{\partial z} - w\frac{\partial}{\partial w}$ on $\mathbb{C}^2$ and let us take the foliation $\widetilde{\mathcal{F}}$ on the blow-up of $\mathbb{C}^2$ at $(0,0)$. We have seen previously that $\widetilde{\mathcal{F}}$ can be locally defined as the kernel of the holomorphic 1-form $\widetilde{\omega} = 2tdz + zdt$ and that $a(p) = l(p) = 1$, thus the exceptional divisor E is $\mathcal{F}$-invariant. We can now calculate $Z(\widetilde{\mathcal{F}}, E) = l(p) + 1 = 2$; the canonical bundle of the foliation $\widetilde{\mathcal{F}}$ is given by

$$K_{\widetilde{\mathcal{F}}} = T^*_{\widetilde{\mathcal{F}}} = \pi^*(T^*_{\mathcal{F}}) \otimes \mathcal{O}_{\widetilde{X}}((1 - l(p))E) = \pi^*(K_{\mathcal{F}})$$

Example 2.2.3. Let X be a compact connected surface and let $\pi : X \to C$ be a fibration, that is a holomorphic surjective morphism onto a curve C with connected fibers. The generic fiber of π is therefore a curve of genus g. We can introduce the relative canonical bundle by the formula:

$$K_{X/C} = K_X \otimes \pi^*(K_C^*).$$

The fibration π can be thought as a foliation $\mathcal{F}$ on X. Due to the fact that we may have ramification on some fibers, we need to pay attention on how we define the conormal bundle $N^*_{\mathcal{F}}$ of the foliation $\mathcal{F}$: given an irreducible component D of a fiber of π, we can denote by l_D the ramification order of π along D. Let η be a local non-vanishing 1-form on the base C; then, $\pi^*\eta$ is by definition a local non-vanishing section of π^*K_C, but also a local section of $N^*_{\mathcal{F}}$ which vanishes on D at order $l_D - 1$, therefore

$$N^*_{\mathcal{F}} = \pi^*(K_C) \otimes \mathcal{O}\Big(\sum (l_D - 1)D\Big).$$

Since $K_X = N^*_{\mathcal{F}} \otimes T^*_{\mathcal{F}}$, we have that the above formula translates into

$$T^*_{\mathcal{F}} = K_X \otimes N_{\mathcal{F}} = K_{X/C} \otimes \pi^*(K_C) \otimes \pi^*(K_C^*) \otimes \mathcal{O}_X\Big(\sum (1 - l_D)D\Big) =$$

$$= K_{X/C} \otimes \mathcal{O}_X\Big(\sum (1 - l_D)D\Big).$$

2.3 Closed positive currents

In this section, we establish an important inequality involving the intersection of the normal bundle with invariant closed positive currents. These results will be especially useful in the classification of non-abundant foliations at the end of Section 3 and in Section 4. Let us begin with a few definitions; all these results can be found in [6], chapter III, unless stated otherwise.

Definition 2.3.1. *Let $G \subseteq \mathbb{R}^n$; then, an upper semi-continuous function $\varphi : G \to \mathbb{R} \cup \{-\infty\}$ is a subharmonic function if, for all closed balls $\overline{B_r(x)} \subset G$ centered in a point $x \in G$ and of radius r, and for every function $h : \overline{B_r(x)} \to \mathbb{R}$ continuous and harmonic in $B_r(x)$ such that $\varphi(y) \leq h(y)$ for all $y \in \partial B_r(x)$, then $\varphi(y) \leq h(y)$ for all $y \in B_r(x)$.*

Definition 2.3.2. *Let $G \subset \mathbb{C}^n$; $f : G \to \mathbb{R} \cup \{-\infty\}$ is a plurisubharmonic function if it is an upper semi-continuous function and for every complex line $\{a + bz : z \in \mathbb{C}\} \subset \mathbb{C}^n$, $a, b \in \mathbb{C}^n$, we have that $z \mapsto f(a + bz)$ subharmonic function on the set $\{z \in \mathbb{C} | a + bz \in G\}$.*

We now give a characterization of plurisubharmonic function which will be useful in Chapter 4:

Lemma 2.3.1. *Let $G \subset \mathbb{C}^n$; $f : G \to \mathbb{R} \cup \{-\infty\}$ is a plurisubharmonic function if and only if for every $a \in G$, for every $\zeta \in \mathbb{C}^n$ such that $|\zeta| < d(a, \partial G)$, then*

$$f(a) \leq \frac{1}{2\pi} \int_0^{2\pi} f(a + \zeta e^{i\theta}) d\theta.$$

Definition 2.3.3. *Given a plurisubharmonic function f, its Lelong number in $x \in \mathbb{C}^n$ is*

$$\nu(f, x) = \liminf_{z \longrightarrow x} \frac{f(z)}{\log |z - x|}.$$

Definition 2.3.4.

- *Let X be a complex manifold of dimension n. Then, a closed positive current Φ is a $(n-p, n-p)$ positive closed form.*
- *Let $\mathcal{F}$ be a foliation on a surface X and let Φ be a $(1,1)$-closed positive current on X with compact support K. Then, Φ is invariant by $\mathcal{F}$ if $\Phi(\beta) = 0$ for every closed 2-form β that vanishes on $\mathcal{F}$.*

Given a closed positive $(1,1)$-current, it can be written as $T = dd^c f$, where f is a psh function and d^c is the operator defined by

$$d^c = \frac{i}{4\pi}(\overline{\partial} - \partial).$$

We can thus define the Lelong number of Φ as

$$\nu(\Phi, x) = \nu(f, x).$$

Let $p \in X \backslash Sing(\mathcal{F})$. Then, in a neighborhood of p we can take local coordinates (z, w) such that $\mathcal{F}$ is given by dz. In this case, β vanishes on $\mathcal{F}$ if and only if $\beta = \beta_1 \wedge dz + \beta_2 \wedge d\overline{z}$ for suitable 1-forms β_1, β_2; this means that Φ is $\mathcal{F}$-invariant if and only if $\Phi = A \cdot dz \wedge d\overline{z}$, where A is a distribution. Since Φ is closed and thus $d\Phi = 0$, A is invariant along the leaves of $\mathcal{F}$, and therefore $A_w = A_{\overline{w}} = 0$; this condition is expressed by saying that Φ is a *transversely invariant measure*.

Let us consider the case where Φ is *diffuse*, i.e. $\nu(\Phi, p) = 0$ for every $p \in X \backslash Sing(\mathcal{F})$. It can still happen the $\nu(\Phi, p) = 0$ for some $p \in Sing(\mathcal{F})$. We shall use the following results that can be found in [8]:

1. the cohomology class $[\Phi]$ of Φ is nef, i.e. $\Phi \cdot C \geq 0$ for every curve $C \subset X$;
2. $[\Phi] \cdot [\Phi] \geq 0$, and $[\Phi] \cdot [\Phi] > 0$ if there exists $p \in X$ such that $\nu(\Phi, p) > 0$.

Then, we can state the following

Theorem 2.3.1. *[3, Theorem 3.5] Let $\mathcal{F}$ be a foliation on a surface X and let Φ be a closed positive current on X with compact support K and invariant by $\mathcal{F}$, and suppose that Φ is diffuse. Then,*

$$N_{\mathcal{F}} \cdot [\Phi] \geq 0$$

and moreover

$$N_{\mathcal{F}} \cdot [\Phi] = 0 \quad \text{if} \quad [\Phi] \cdot [\Phi] = 0.$$

2.4 Hirzebruch-Jung Singularities

In this paragraph, we introduce a special kind of singularity of a surface which will be useful both for some examples, but also for the next section. The reference for this chapter is [1, Chapter III.5].

Definition 2.4.1. *Let r, b be coprime integers with $r > b > 0$. Then, the Hirzebruch-Jung continued fraction of $\frac{r}{b}$ is the expression*

$$\frac{r}{b} = a_1 - \frac{1}{a_2 - \frac{1}{a_3 - \dots}} = [a_1, a_2, ..., a_k].$$

Let now $C = \sum_{i=1}^{r} C_i$ be a curve on the surface X, such that each C_i is a smooth rational curve, $C_i^2 = e_i \leq -2$, $C_i C_j = 1$ for $|i-j| = 1$ and $C_i C_j = 0$ for $|i-j| > 1$. This is called a *Hirzebruch-Jung string*. For a sufficiently small X, we can find a smooth curve C_0 which intersects C_1 transversely at one point and does not intersect any of the curves $C_2, ..., C_r$; similarly, we can find a smooth curve C_{r+1} that only intersects C_r in one point and is disjoint from all the other curves. We want to find some indices $n_i, i = 0, ..., k+1$, such that we can find a holomorphic function φ on X with divisor $(\varphi) = \sum_0^{r+1} n_i C_i$; this can be done if and only if

$$n_{k-1} + e_k n_k + n_{k+1} = \sum_{i=0}^{r+1} n_i C_i C_k = 0$$

therefore, given n_0 and n_1 the coefficients $n_k, k = 2, ..., r+1$ can be recursively determined by the formula

$$n_k = |e_{k-1}| \cdot n_{k-1} - n_{k-2}.$$

If $n_0 \leq n_1$, then $n_k \leq n_{k+1}$ for all k, so if we determine integers μ_k, ν_k starting with initial data $\mu_0 = 0, \mu_1 = 1, \nu_0 = 0, \nu_1 = 1$, we have holomorphic functions g, h on X with divisors

$$(g) = \sum_{i=0}^{r+1} \mu_i C_i, \quad (h) = \sum_{i=0}^{r+1} \nu_i C_i.$$

The integers μ_i satisfy

$$\frac{\mu_{k+1}}{\mu_k} = \big[|e_k|, |e_{k-1}|, ..., |e_1|\big];$$

in particular, if $n' = \mu_{r+1}, q' = \mu_r$, these are still coprime and the expression

$$\frac{n'}{q'} = [|e_r|, |e_{r-1}|, ..., |e_1|]$$

shows that the e_i are determined by n', q'. Finally, we define a divisor

$$(f) = \sum_{i=0}^{r+1} \lambda_i C_i$$

by integers λ_i satisfying the recursion formula and $\lambda_{r+1} = 0, \lambda_r = 1$. Hence, the λ's are the μ's that we would have obtained by starting with our index at the other end of the string, so $\lambda_1 = q, \lambda_0 = n$ and so $n = n'$. We can prove that, for the functions f, g, h, we have the relation $h^n = fg^{n-q}$; therefore, a change of coordinates

$$w = h, z = f, u = g$$

maps X into the surface

$$W = \big\{(w, z, u) \in \mathbb{C}^3 : w^n = zu^{n-q}\big\}.$$

Theorem 2.4.1. *[1, Theorem 5.1] For $0 < q < n$, n and q coprime, let $C \subset X$ be the Hirzebruch-Jung string with self-intersection numbers e_i satisfying $\frac{n}{q} = \big[|e_1|, |e_2|, ..., |e_r|\big]$, and let $y \in Y$ be the singularity resulting from contracting C. Then, this singularity is isomorphic to the singularity on $0 \in \mathbb{C}^3$ in the normalization of the surface W above.*

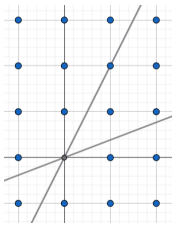
Definition 2.4.2. *This singularity is called an Hirzebruch-Jung singularity, and since it only depends on n and q, we denote it as $A_{n,q}$.*

2.5 Examples of important foliations

In this section, we want to give some examples of important foliations that will either clarify some properties, or be of significant relevance for the classification.

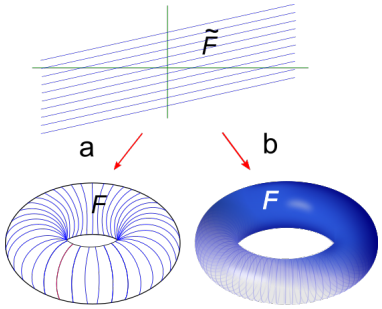
2.5.1 A foliation with non-algebraic leaves

Let $\widetilde{\mathcal{F}}$ be a linear foliation on $\mathbb{C}^2$, generated by the vector field $v = m\frac{\partial}{\partial w} + \frac{\partial}{\partial z}$.



This vector field is invariant by translation, therefore if we quotient by a lattice Λ we obtain a well-defined vector field on the torus $T = \mathbb{C}^2/\Lambda$, which generates a foliation $\mathcal{F}$.

If the leaves of $\widetilde{\mathcal{F}}$ pass by the points of the lattice, then all leaves of $\mathcal{F}$ are closed curves homeomorphic to the circle; otherwise, they are non-compact and dense in the torus, and they are not algebraic.



source: Wikipedia

2.5.2 Riccati foliations

The reference for this section is [3, Section 4.1].

Definition 2.5.1. *A foliation $\mathcal{F}$ on a compact connected surface X is called Riccati foliation if there exists a rational fibration $\pi : X \to B$, possibly with singular fibers, whose generic fiber is transverse to $\mathcal{F}$. The fibration π is called adapted fibration, or we say that $\mathcal{F}$ is Riccati with respect to π.*

Let $\pi : X \to B$ be a rational fibration on X and let F be a regular fiber of π, therefore a smooth rational curve of self-intersection 0. Let $\mathcal{F}$ be a foliation on X with tangent bundle $T_{\mathcal{F}}$ and normal bundle $N_{\mathcal{F}}$; then $\mathcal{F}$ is Riccati with respect to π if and only if $T_{\mathcal{F}} \cdot F = 0$, or equivalently by adjunction $N_{\mathcal{F}} \cdot F = 2$.

Proof. Let $\mathcal{F}$ be Riccati with respect to π; then, if F is a fiber transverse to $\mathcal{F}$, we have $T_{\mathcal{F}} \cdot F = F \cdot F = 0$ and this holds for any regular fiber since they all are in the same cohomology class.

Conversely, if $T_{\mathcal{F}} \cdot F = 0$, then from the adjunction formulae we get that either F is not $\mathcal{F}$-invariant and $tang(\mathcal{F}, F) = 0$, and this means F is transverse to $\mathcal{F}$, or F is $\mathcal{F}$-invariant and $Z(\mathcal{F}, F) = 2$. But a positive $Z(\mathcal{F}, F)$ means that $Sing(\mathcal{F}) \cap F \neq \emptyset$, and we only have a finite number of singular points; therefore, the second case can only happen for a finite number of fibers. Hence, $\mathcal{F}$ is Riccati with respect to π. □

If $T_{\mathcal{F}} \cdot F = 2$, or equivalently $N_{\mathcal{F}} \cdot F = 0$, then $\mathcal{F}$ coincides with the rational fibration: in fact, if we had a regular fiber F of π that is not $\mathcal{F}$-invariant, we would obtain from Proposition 2.2.1 $tang(\mathcal{F}, F) = -2 < 0$, which is impossible. It also cannot happen that $T_{\mathcal{F}} \cdot F = 1$: if that were the case, we would again have $tang(\mathcal{F}, F) = -1 < 0$. Since (from [1, p.142]) we know that if F is a

smooth rational curve of zero self-intersection on a compact connected surface X, then F is a regular fiber of a rational fibration X, we immediately deduce that the following proposition holds:

Proposition 2.5.1. *[3, Prop. 4.1] Let $\mathcal{F}$ be a foliation on a compact connected surface X and let $F \subset X$ be a smooth rational curve of zero self-intersection and invariant by $\mathcal{F}$. Then:*

1. *if $Z(\mathcal{F}, F) = 0$, then $\mathcal{F}$ is a rational fibration;*
2. *if $Z(\mathcal{F}, F) = 2$, then $\mathcal{F}$ is a Riccati foliation;*
3. *the case $Z(\mathcal{F}, F) = 1$ cannot happen.*

On a neighborhood of each transverse fiber, a Riccati foliation $\mathcal{F}$ defines through its leaves a local trivialization of the adapted fibration $\pi : X \to B$, and thus a canonical identification between fibers closed to the adapted fibration. Glueing together these local identifications, we obtain a group representation, namely the *monodromy of $\mathcal{F}$*:

$$\rho : \pi_1(B^*, b) \longrightarrow Aut(\pi^{-1}(b)),$$

where $B^* \subset B$ is the set of fibers transverse to $\mathcal{F}$, b is a point of B^*, $\pi_1(B^*, b)$ is the fundamental group, and $Aut(\pi^{-1}(b))$ is the group of holomorphic automorphism of the fiber $\pi^{-1}(b) \simeq \mathbb{C}P^1$. We shall often refer to $\mathcal{F}\big|_{X^*}$, where $X^* = \pi^{-1}(B^*)$, as the *suspension of the representation ρ*. The usefulness of the representation lies in the fact that the dynamics of $\mathcal{F}\big|_{X^*}$ is quite well represented by the dynamics of $Im(\rho) \subset Aut(\mathbb{C}P^1)$: for example, dense leaves of $\mathcal{F}\big|_{X^*}$ correspond to dense orbits of ρ. When $B^* = B$, i.e. each fiber is transverse to the foliation, the monodromy ρ and the analytic type of the base B completely characterize $\mathcal{F}$ up to biholomorphism.

2.5.3 A very special foliation

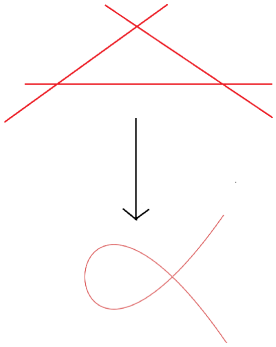
The reference for this section is [3, Section 4.2]

Let T be an automorphism of $\mathbb{C}P^2$ that cyclically permutes 3 non-collinear points p_1, p_2, p_3. In suitable projective coordinates, $T : (s : t : u) \mapsto (u : s : t)$; we have that $T^3 = id$ and T has three fixed points q_1, q_2, q_3 located at $(1 : \beta : \beta^2)$, where β is a cubic root of 1. We want to find a foliation that is invariant by T: let $\mathcal{L}$ be a foliation generated by a linear vector field with zeroes in p_1, p_2, p_3, that is $v = \lambda_1 z\frac{\partial}{\partial z} + \lambda_2 w\frac{\partial}{\partial w}$, where $z = \frac{s}{u}, w = \frac{t}{u}$. By calculations, we can prove that the only linear T-invariant foliations are those generated by the vector fields

$$z\frac{\partial}{\partial z} + \frac{1}{2}(1 \pm i\sqrt{3})w\frac{\partial}{\partial w}.$$

Consider now the quotient surface $Y_0 = \frac{\mathbb{C}P^2}{T}$. $\mathcal{L}$ is invariant by T, so we have on Y_0 a quotient foliation $\mathcal{H}_0 = \frac{\mathcal{L}}{T}$. Y_0 is a singular surface: it has three singular points $\hat{q}_1, \hat{q}_2, \hat{q}_3$ in correspondence with the three fixed point q_1, q_2, q_3 of T. These

three points are of type $A_{3,2}$, therefore the minimal resolution of each one of them consists of two (-2)-curves D_j, E_j intersecting at one point. Let Y be the minimal resolution of Y_0 and $\mathcal{H}$ the lift of $\mathcal{H}_0$ on Y. Then the structure of $\mathcal{H}$ around the curves D_j, E_j is easily understood, because the original $\mathcal{L}$ is not singular at q_j, thus D_j, E_j are both $\mathcal{H}$-invariant. Besides the curves E_j, D_j, $\mathcal{H}$ has another invariant curve, which comes from the quotient of the three lines of $\mathbb{C}P^2$ through p_1, p_2, p_3. These lines are cyclically permuted by T, therefore their quotient on Y_0 is a rational curve with a node. This curve is disjoint from the other singularities of Y_0 and its lift to Y will be noted as C. It is a rational curve with a node p, with $C \cdot C = 3$, invariant by $\mathcal{H}$.



The singularity of $\mathcal{H}$ at p is the same as the singularities of $\mathcal{L}$ at p_j, i.e a reduced non-degenerate singularity with eigenvalue $\lambda = \frac{1+i\sqrt{3}}{2}$. There are no more $\mathcal{H}$-invariant curves since the three lines above are the only ones invariant by $\mathcal{L}$, due to the fact that λ is not rational; moreover, there are no more singularities besides the ones on C, E_j, D_j.

Let us observe that, if we blow up at $p \in C$, then the strict transform of $\widetilde{C}$ of C becomes a (-1)-curve, intersecting the exceptional divisor at two points; then, we can contract $\widetilde{C}$ and obtain a foliation $\mathcal{H}'$ which is holomorphic to $\mathcal{H}$. Hence, $\mathcal{H}$ possesses some non-trivial birational automorphism and this fact will prove useful in the next chapter.

We can now state a result that can be found in [3, Proposition 4.3] that provides a characterization of $\mathcal{H}$ in terms of the curve C.

Proposition 2.5.2. *Let $\mathcal{F}$ be a foliation on a compact connected surface X and let $C \subset X$ be a rational curve with a node p, invariant by $\mathcal{F}$, and with $C \cdot C = 3$. Suppose that p is a reduced non-degenerate singularity of $\mathcal{F}$, and that it is the unique singularity of $\mathcal{F}$ on C. Then $\mathcal{F}$ is birational to $\mathcal{H}$.*

It is also important to state that

Proposition 2.5.3. *[3, Proposition 4.4] $\mathcal{H}$ is not birational to a Riccati foliation.*

2.5.4 Turbulent foliations

The reference for this section is [3, Section 4.3].

Definition 2.5.2. *A foliation $\mathcal{F}$ on a compact connected surface X is called turbulent if there exists an elliptic fibration $\pi : X \to B$ whose generic fiber is transverse to $\mathcal{F}$.*

Since the generic fiber F of π is elliptic, we have $\chi(F) = 0$, and since it is a fiber, $F \cdot F = 0$. Therefore, the condition $T_{\mathcal{F}} \cdot F = 0$, or equivalently $N_{\mathcal{F}} \cdot F = 0$, means that either $\mathcal{F}$ is turbulent with respect to π or that $\mathcal{F}$ coincides with π.

Proposition 2.5.4. *Let $\mathcal{F}$ be a foliation on a compact connected surface X and let $F \subset X$ be a regular fiber of an elliptic fibration π on X, invariant by $\mathcal{F}$ and with $Z(\mathcal{F}, F) = 0$. Then, either $\mathcal{F}$ is a turbulent foliation with respect to π, or it coincides with the elliptic fibration π.*

The difference with Proposition 2.5.1 is that we have to explicitly assume that F is a fiber of an elliptic fibration: in general, a smooth elliptic curve of zero self-intersection might not have such a property, as we can see in

Example 2.5.1. Let $X = \frac{\mathbb{C}^* \times \mathbb{C}P^1}{(z,w)\sim(2z,w+1)}$, and let $\mathcal{F}$ be the foliation on X given by the quotient of $\{dw = 0\}$. Then, $\mathcal{F}$ is tangent to the elliptic curve $F \subset X$ which corresponds to $\{w = \infty\}$ and is such that $F \cdot F = 0$ and $\mathcal{O}_X^*(F)\big|_F = \mathcal{O}_F$, but is not a fiber of an elliptic fibration.

As in the Riccati case, given a turbulent foliation $\mathcal{F}$ with adapted fibration $\pi : X \to B$, we can consider its monodromy

$$\rho : \pi_1(B^*, b) \to Aut(\pi^{-1}(b)),$$

where $B^* \subset B$ is te set of fibers transverse to $\mathcal{F}$ and $b \in B^*$. Let us observe that all the fibers $\pi^{-1}(b), b \in B^*$, are isomorphic to the same elliptic curve E, where the isomorphism is given by the foliation; this means that $X^* = \pi^{-1}(B^*)$ is actually an elliptic fiber bundle with fiber E.

2.6 Foliations generated by global holomorphic vector fields

In this section, we see some results about foliations generated by global holomorphic vector fields; the results will be particularly meaningful once we start dealing with the classification of foliations on surfaces. We will start with a quick overview of some definitions and results on the Albanese morphism, and then we will move on following [3, Section 6] to define and describe logarithmic foliations.

Let X be a compact connected surface; then, we know from Hodge Theory that $H^{1,0}(X) = H^0(X, \Omega_X^1) \cong \mathbb{C}^q$. We can also construct an injective morphism

$$\begin{aligned} H_1(X, \mathbb{Z}) &\longrightarrow H^0(X, \Omega_X^1) \\ \gamma &\longmapsto \left(\omega \mapsto \int_\gamma \omega\right). \end{aligned}$$

We know from algebraic topology results that $H_1(X,\mathbb{Z}) \cong \mathbb{Z}^{2q}$. Thanks to the morphism above, we can see $H_1(X,\mathbb{Z})$ as a soubgroup of $H^0(X,\Omega^1_X)$. Therefore, the quotient

$$Alb(X) = \frac{H^0(X,\Omega^1_X)^*}{H_1(X,\mathbb{Z})}$$

is well defined and it is a torus, called the *Albanese torus* of X. If we fix a point $p \in X$, we can also define the *albanese morphism*

$$\begin{aligned} alb_X : X &\longrightarrow Alb(X) \\ x &\mapsto \left(\omega \mapsto \int_{\gamma_{px}} \omega\right). \end{aligned}$$

By construction, this map has the property that

$$alb_X^* : \Omega^1_{Alb(X)}(Alb(X)) \longrightarrow \Omega^1_X(X)$$

is an isomorphism.

The only holomorphic 1-forms on a torus are the linear ones, which we encountered before, and they generate a class of foliations, called *Kroenecker foliations*, whose leaves are traslations of a codimension one subgroup of $Alb(X)$. In this way, we see that the foliations on X generated by a global holomorphic 1-form are the pull-back by alb_X of a Kroenecker foliation on $Alb(X)$.

If the image of alb_X is a curve, then alb_X is a fibration over that curve, the so-called *Albanese fibration*. This is now the only foliation on X with effective conormal bundle. We shall use this fact in the equivalent form:

Remark 2.6.1. If X is a surface such that $\Omega^1_X(X)$ is non-empty and $\mathcal{F}$ is a foliation on X such that $h^0(X,T^*_{\mathcal{F}}) = 0$, then $\mathcal{F}$ is the Albanese fibration.

We can now state a result due to Castelnuovo and de Franchis:

Theorem 2.6.1 (Castelnuovo-de Franchis)**.** *[3, Proposition 6.1] Let X be a compact connected surface and let $\mathcal{F}$ be a foliation on X such that $h^0(X,N^*_{\mathcal{F}}) \geq 2$. Then, $\mathcal{F}$ is a fibration over a curve of genus $g \geq 2$.*

Let now X be a connected surface and $C \subset X$ be a curve.

Definition 2.6.1. *A logarithmic 1-form on X with poles on C is a meromorphic 1-form on $U \subset X$ such that for every $p \in U$, there exists $V \subset U$, $p \in V$, such that*

$$\omega\big|_V = \omega_0 + \sum_{j=1}^{n} g_j \frac{df_j}{f_j}$$

where ω_0 is a holomorphic 1-form on V, f_j, g_j are holomorphic functions on V for all j, $\{f_j = 0\}$, $j = 1,...,n$ reduced equation of the irreducible components of $C \cap V$.

Let now $\mathcal{F}$ be a foliation on X and let C be an $\mathcal{F}$-invariant curve. On a neighborhood V of a point $p \in C$, take a 1-form ω, holomorphic and with isolated zeroes, and a reduced equation f on $C \cap V$. Then, the meromorphic 1-form $\frac{\omega}{f}$ has first order poles on C; if it is also logarithmic around p, then we say that the foliation $\mathcal{F}$ is logarithmic at p along C. If this property holds for all $p \in C$, then $\mathcal{F}$ is a logarithmic foliation on (X, C). Conversely, any logarithmic 1-form on a surface X, with empty zero divisor and polar divisor equal to C defines a logarithmic foliation on (X, C).

Proposition 2.6.1. *[3, Prop. 6.4] Let X be a compact connected surface and let $C \subset X$ be a curve, possibly empty. Let $\mathcal{F}$ be a logarithmic foliation on (X, C). Then,*

$$\mathrm{kod}(N_{\mathcal{F}}^* \otimes \mathcal{O}_X^*(C)) \leq 1$$

and if $\mathrm{kod}(N_{\mathcal{F}}^* \otimes \mathcal{O}_X^*(C)) = 1$, then $\mathcal{F}$ is a fibration.

We shall recall in Section 3 the definitions and basic properties of of the Kodaira dimension of a line bundle, which appears in this statement; for now, the only property that concerns us is that $\mathrm{kod}(L) \geq 1$ means that some positive tensor power of L has two linearly independent global sections.

Let X be a compact connected surface and let v be a non-trivial holomorphic vector field on X. If D denotes its zero divisor, and if $\mathcal{F}$ is the foliation generated by v, then $T_{\mathcal{F}} = \mathcal{O}_X^*(D)$ and $N_{\mathcal{F}}^* = K_X \otimes \mathcal{O}_X^*(D)$. Hence, X cannot be a surface of general type, i.e. $\mathrm{kod}(X) \leq 1$: otherwise, we would contradict Bogomolov's Lemma. By the same lemma, if $\mathrm{kod}(X) = 1$, then $\mathrm{kod}(N_{\mathcal{F}}^*) = 1$ and so $\mathcal{F}$ is a fibration over a curve B. The generic fibers are curves with a non-trivial vector field, hence rational or elliptic, but the fact that $\mathrm{kod}(X) = 1$ means that the fibers are actually elliptic. Thus, $\mathcal{F}$ is an elliptic fibration.

Lemma 2.6.1. *[3, Lemma 6.1] Let $\pi : X \to B$ be an elliptic fibration, and let v be a non-trivial holomorphic vector field tangent to the fibers of π. Then, every fiber of π is a smooth elliptic curve (but possibly multiple), and outside multiple fibers π is a locally trivial fibration. Moreover, v has empty zero set.*

In this case, we will call π or $\mathcal{F}$ an *almost elliptic fiber bundle*.

Suppose now that $\mathrm{kod}(X) = 0$ and that X is minimal, i.e. there are no (-1)-curves on X. This means that there exists a positive integer m such that $K_X^{\otimes m}$ is trivial. Then, there exists regular covering $Y \to X$ such that $K_Y = \mathcal{O}_Y$. The vector fields on X lift to vector fields on Y, and since the canonical bundle of Y is trivial, vector fields on Y correspond to 1-forms on Y. The type of the foliations on Y is thus completely described by the dimension of the image of the Albanese morphism alb_Y: indeed, if the image is one-dimensional, then any holomorphic vector field on Y is tangent to the fibers of the Albanese fibration $Y \to B$, which will be an almost elliptic fiber bundle by the lemma above. If the dimension of the image of alb_Y is two, then there are on Y two holomorphic vector fields v_1, v_2 with $v_1 \wedge v_2 \not\equiv 0$, and since K_Y^* is trivial, v_1, v_2 are everywhere linearly independent and commute. We deduce that Y is a torus and the only

global holomorphic vector fields on Y are the linear ones, generating Kroenecker foliations.

In both cases, when we quotient on X we get an almost elliptic fiber bundle. We can resume the results just obtained in the following

Proposition 2.6.2. *[3, Prop. 6.5] Let X be a connected compact surface of non-negative Kodaira dimension and let v be a non-trivial holomorphic vector field on X. Then, either v generates an almost elliptic fiber bundle, or it generates a Kroenecker foliation on a torus.*

If $\mathrm{kod}(X) = -\infty$ and X is algebraic, then we have the following result

Proposition 2.6.3. *[3, Prop. 6.6] Let X be an algebraic surface such that $\mathrm{kod}(X) = -\infty$ and let v be a non-trivial holomorphic vector field on X. Then,*

1. *if $h^{1,0}(X) \geq 2$, then v is tangent to the Albanese fibration;*
2. *if $h^{1,0}(X) = 1$, then either v is tangent to the Albanese fibration, or it is the suspension of a representation $\rho : \pi_1(Alb(X)) \to Aut(\mathbb{C}P^1)$;*
3. *if $h^{1,0}(X) = 0$, then there exists a birational map $X \dashrightarrow \mathbb{C}P^1 \times \mathbb{C}P^1$ sending v to a holomorphic vector field $v_1 \oplus v_2$ on $\mathbb{C}P^1 \times \mathbb{C}P^1$.*

3 A first classification of foliations

3.1 Birational classification of surfaces

In this section, we want to present some meaningful results about the birational classification of surfaces, in order to be able to make a comparison with the birational classification of holomorphic foliations on surfaces. In particular, we will briefly present Mori's program, whose intent is to classify surfaces by making the canonical bundle K_X "as positive or negative as possible"; we will see later on how can this be adapted to the context of foliations on surfaces.

Definition 3.1.1. *Let X be a smooth projective surface; we say that a curve $E \subset X$ is a (-1)-curve if $E \simeq \mathbb{C}P^1$, $E^2 = -1$.*

Theorem 3.1.1 (Castelnuovo's contractibility theorem)**.** *Let $\widetilde{X}$ be a smooth projective surface, let $E \subset \widetilde{X}$ be a (-1)-curve; then, there exists a smooth algebraic surface X and a morphism β s.t.*

$$\begin{aligned} \beta : \widetilde{X} &\longrightarrow X \\ E &\longmapsto p \end{aligned}$$

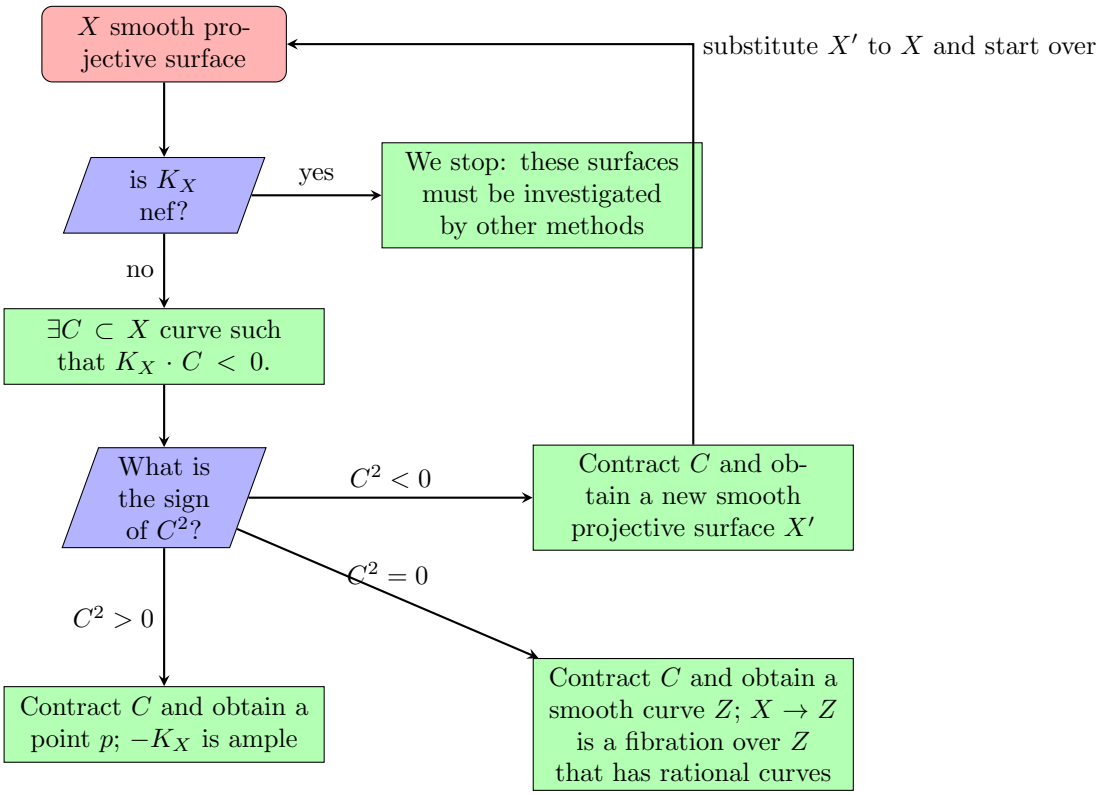
and $\beta\big|_{\beta^{-1}(X\setminus p)}$ is an isomorphism.

Given a smooth projective surface X, we define the set $\mathcal{B}(X) = \{\text{isomorphism classes of surfaces birational to } X\}$. We write $X_1 \geq X_2$ if there exists a birational morphism $X_1 \longrightarrow X_2$; this is an order relationship. We say that a *minimal model* of X is a surface $X_0 \in \mathcal{B}(X)$ such that X_0 is minimal with respect to $\geq$. X_0 is called a *minimal surface*. We can prove a useful characterization of minimal surfaces:

Lemma 3.1.1. *A surface X is minimal if and only if there are no (-1)-curves in it.*

With this in mind, it is also not so difficult to prove that every surface admits a minimal model. We know from Enriques-Kodaira classification that this minimal model is unique if and only if the Kodaira dimension of X is non-negative.

We follow this step-by-step approach to the minimal model of surfaces (all the details can be found in [13]):



3.2 Minimal models

The reference for all the theorems in this section is [3], unless stated otherwise. From now on, we will use the notation $(X, \mathcal{F})$ to denote a foliation $\mathcal{F}$ on a compact connected surface X, and $(X, \mathcal{F}) \dashrightarrow (Y, \mathcal{G})$ to denote a birational map from X to Y sending $\mathcal{F}$ to $\mathcal{G}$.

As we have stated before, in order to study the foliations modulo birationality we need to work with reduced singularities. Thanks to Seidenberg theorem, we know that any foliation is birational to one with reduced singularities; however, such a foliation is never unique since we can blow up any point to obtain a new foliation with reduced singularities, birational to the one we started with. For this reason, we want to give some definition that can provide us with a better concept of "unicity", while also defining the analogous in foliation theory of (-1)-curves and minimal models.

Definition 3.2.1. *Given a foliation $(X, \mathcal{F})$, we say that a curve $C \subset X$ is $\mathcal{F}$-exceptional if:*

1. *C is an exceptional curve, i.e. a smooth rational curve of self intersection -1;*

2. *the contraction of C to a point p gives a new foliation $(X', \mathcal{F}')$ which has at p a regular point or a reduced singular point.*

Definition 3.2.2. *A foliation $(X, \mathcal{F})$ is said to be relatively minimal if*

1. *it is reduced;*

2. *there are no $\mathcal{F}$-exceptional curves.*

In order to study $\mathcal{F}$-exceptional curves, we can use the propositions and formulae that we have seen in the previous section. If C is an $\mathcal{F}$-exceptional curve, then by definition, its contraction is either a regular or a reduced singular point. The fact that the singularity is not dicritical means that C is an $\mathcal{F}$-invariant curve from Proposition 2.1.1; there are two possibilities:

1. C is $\mathcal{F}$-invariant and contains one singularity q of $\mathcal{F}$, with $Z(\mathcal{F}, C, q) = 1$; by contracting C we obtain a regular point p of the contracted foliation $\mathcal{F}'$, since its order $a(p) = 0$: we have seen before that in the invariant case, $Z(\mathcal{F}, C) = l(p) + 1$ and $a(p) = l(p)$.

2. C is $\mathcal{F}$-invariant and contains two singularities q_1, q_2 of $\mathcal{F}$, which are reduced and satisfy $Z(\mathcal{F}, C, q_1) = Z(\mathcal{F}, C, q_2) = 1$; by contracting C we obtain a singular point of the foliation $\mathcal{F}'$, which is a reduced singularity since $a(p) = 1$ and the eigenvalues of its linear part have no positive rational quotient. Indeed, since q_1, q_2 are reduced singularities and those singularities can be degenerate or non-degenerate, we have three cases; by computing explicitly the blow-down maps we find that if q_1, q_2 are both reduced non-degenerate, then p is also reduced non-degenerate, while if one of the two is a saddle node, then p is a saddle-node aswell. Lastly, the same computation shows that q_1, q_2 cannot both be saddle-nodes.

   Let us also observe that, in this case, we cannot replace the condition $Z(\mathcal{F}, C, q_1) = Z(\mathcal{F}, C, q_2) = 1$ with $Z(\mathcal{F}, C) = 2$, as the next example shows:

   Example 3.2.1. Let $(\mathcal{F}, \mathbb{C}^2)$ be the foliation generated by the 1-form $\omega = (z+w)dw - wdz$, i.e. by the vector field $v = (z+w)\frac{\partial}{\partial z} + w\frac{\partial}{\partial w}$; then, $(0,0)$ is a singular point. The linear part of the vector field is $(Dv)((0,0)) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, therefore $\lambda = 1$ and the singularity is non-reduced. If we blow up in $(0,0)$, in local coordinates around $((0,0),(0:1))$ we find

$$\pi_0 : \widetilde{U}_0 \longrightarrow \mathbb{C}P^2$$
$$(z,t) \longmapsto (z, zt),$$

   so $\pi_0^*\omega = (z + zt)d(zt) - ztdz = z[z(1+t)dt + t^2dz] = z\widetilde{\omega}_0$, and this means that $l(p) = 1 = a(p)$ and the exceptional divisor $E = \{z = 0\}$ is $\widetilde{\mathcal{F}}$-invariant. In the other chart, we get

$$\pi_1 : \widetilde{U}_1 \longrightarrow \mathbb{C}P^2$$
$$(s,w) \longmapsto (sw, w),$$

thus $\pi_1^*\omega = w[dw - wds] = w\widetilde{\omega}_1$, and $\widetilde{\omega}_1$ is non-singular. $l(p) = a(p) = 1$ in this case as well. From [3], p.17, we have

$$K_{\widetilde{\mathcal{F}}} = \pi^* K_{\mathcal{F}} \otimes \mathcal{O}_{\widetilde{X}}((1 - l(p))E) = \pi^* K_{\mathcal{F}},$$

thus p is not a reduced singularity, but it is canonical in the sense of [3], Definition 8.2.

Since E is an exceptional curve, it is a smooth rational curve of self-intersection -1; moreover, $Z(\widetilde{\mathcal{F}}, E) = l(p) + 1 = 2$, but E is not an $\mathcal{F}$-exceptional curve since its contraction gives a non-reduced singularity.

Proposition 3.2.1. *Any foliation $(X, \mathcal{F})$ has a relatively minimal model.*

Proof. By Seidenberg's Theorem, there exists a foliation with reduced singularities $(\widetilde{X}, \widetilde{\mathcal{F}})$ and a birational morphism $(\widetilde{X}, \widetilde{\mathcal{F}}) \to (X, \mathcal{F})$. If this is relatively minimal, then we are done; otherwise we can contract one by one all the $\mathcal{F}$-exceptional curves, and this process must end because every time we iterate, we reduce by one the rank of the Néron-Severi group of the surface. Once there are no more $\mathcal{F}$-exceptional curves, we are left with a relatively minimal foliation, birational to $(X, \mathcal{F})$. □

We would like to understand if such relatively minimal models can be unique in some sense. For this reason, we give the following definition.

Definition 3.2.3. *A foliation $(X, \mathcal{F})$ with reduced singularities is called minimal if*

1. *it is relatively minimal;*
2. *if $(Y, \mathcal{G})$ is another relatively minimal foliation and $f : (X, \mathcal{F}) \dashrightarrow (Y, \mathcal{G})$ is any birational map sending $\mathcal{F}$ to $\mathcal{G}$, then f is in fact a biregular map.*

Remark 3.2.1. In Definition 3.2.3, we do not excude the case where $(Y, \mathcal{G}) = (X, \mathcal{F})$; therefore, a minimal foliation has the property that any birational self-map is, in fact, biregular. This means that if we already know that a relatively minimal foliation has non-trivial birational self-maps, then such foliation is not minimal. This is, for example, the case of the very special foliation $\mathcal{H}$.

Besides the above mentioned very special foliation $\mathcal{H}$, we can find some other examples of foliations without a minimal model:

Example 3.2.2. Let $\mathcal{F}$ be an algebraically integrable foliation induced by a morphism $\pi : X \to B$, which is a fibration in rational curves onto a curve B; then, $\mathcal{F}$ has reduced singularities and the only $\mathcal{F}$-exceptional curves are contained in the singular fibers. However, $\mathcal{F}$ is never minimal, as shown by the flipping of a fiber, i.e. blowing up one point on a fiber and contracting the strict transform of the fiber itself.

Example 3.2.3. Let $\mathcal{F}$ be a Riccati foliation and suppose there exists a regular fiber F of the adapted fibration which is invariant by $\mathcal{F}$ and which contains two reduced singularities. If $\mathcal{F}$ is relatively minimal, then we can take a birational map onto a relatively minimal model $\gamma : \mathcal{F} \dashrightarrow \mathcal{F}_0$ which has no indeterminacy points over F; for example, γ might be the blow-up of a point outside of F, composed with the contraction of a fiber intersecting the exceptional divisor, and then we can contract all $\mathcal{F}$-exceptional curves. In this way, the image of F will be an $\mathcal{F}_0$-invariant curve F_0, showing that $\mathcal{F}_0$ is not minimal.

Now that we have seen some examples, it is in our interest to classify all those foliation that do not present a minimal model. We shall see that the few examples we have seen are in fact the only ones.

Theorem 3.2.1. *Let $(X, \mathcal{F})$ be a foliation that does not admit a minimal model. Then, $(X, \mathcal{F})$ is birational to a foliation in the following list:*

1. *foliation by rational curves;*
2. *non-trivial Riccati foliations;*
3. *the very special foliation $(Y, \mathcal{H})$ of the previous paragraph.*

Proof. Without loss of generality, we may suppose thet $(X, \mathcal{F})$ is relatively minimal. By hypothesis, it is not minimal and therefore there exists a bimeromorphic map $f : (Y, \mathcal{G}) \dashrightarrow (X, \mathcal{F})$, with $(Y, \mathcal{G})$ reduced, which is not a morphism. Let $\pi : \widetilde{Y} \to Y$ be the minimal resolution of the indeterminacies of f, so that $\widetilde{f} : (\widetilde{Y}, \widetilde{\mathcal{G}}) \to (X, \mathcal{F})$ is the induced morphism. Then, $\widetilde{Y}$ contains an exceptional curve C which is not contracted by $\widetilde{f}$ but is contracted by π, and C is $\widetilde{\mathcal{G}}$-exceptional because $\mathcal{G}$ is reduced. $\widetilde{f}$ is not a biholomorphism on a neighborhood of C, otherwise $\widetilde{f}(C)$ would be an $\mathcal{F}$-exceptional curve, contrary to the relative minimality of $(X, \mathcal{F})$. On the other hand, $\widetilde{f}$ is a composition of blow-ups of X, hence can be factorized as $h \circ g : (\widetilde{Y}, \widetilde{\mathcal{G}}) \to (\widetilde{X}, \widetilde{\mathcal{F}}) \to (X, \mathcal{F})$ in such a way that g is a biholomorphism on a neighborhood of C and $C_1 = g(C)$ intersects an exceptional curve C_2 of $\widetilde{X}$ contracted by h. Since $\mathcal{F}$ is reduced, C_1 and C_2 are $\mathcal{F}$-exceptional. We have just stated that, if $(X, \mathcal{F})$ has no minimal model, then it is bimeromorphic to a reduced foliation $(\widetilde{X}, \widetilde{\mathcal{F}})$ which contains two $\widetilde{\mathcal{F}}$-exceptional curves C_1, C_2 with $C_1 \cap C_2 \neq \emptyset$.

Let $p \in C_1 \cap C_2$; then $Z(\widetilde{\mathcal{F}}, C_1, p) = Z(\widetilde{\mathcal{F}}, C_2, p) = 1$ and so p is reduced non-degenerate; in particular, the intersections between the two curves are transverse and their number is at most 2. The proof is completed by the following case-by-case analysis:

1. If $|C_1 \cap C_2| = 1$, then the contraction of C_1 transforms C_2 into C_2', which is a smooth rational curve of zero self-intersection: indeed, $\pi^* C_2' = C_1 + C_2$ and $C_2' \cdot C_2' = (C_1 + C_2)^2 = C_1^2 + C_2^2 + 2C_1 \cdot C_2 = -1 - 1 + 2 = 0$. Moreover, C_2' is invariant by the induced foliation $\overline{\mathcal{F}}$, which is still reduced. The point $q \in C_2'$ under C_1 is either regular for $\overline{\mathcal{F}}$ or singular with $Z(\overline{\mathcal{F}}, C_2', q) = 1$, hence $Z(\overline{\mathcal{F}}, C_2') \leq 2$ and Proposition 2.3.1 allows to conclude that $\overline{\mathcal{F}}$ is either a rational fibration or a Riccati foliation.

2. If $|C_1 \cap C_2| = 2$, then by contracting C_1 the curve C_2 is transformed into a rational curve with a node C_2', of self-intersection 3: indeed, $\pi^* C_2' = 2C_1 + C_2$ and $C_2' \cdot C_2' = (2C_1 + C_2)^2 = 4C_1^2 + C_2^2 + 4C_1 \cdot C_2 = -4 - 1 + 8 = 3$. The induced foliation $\overline{\mathcal{F}}$ satisfies the hypothesis of Proposition 2.3.3, and therefore is bimeromorphic to the very special foliation $(Y, \mathcal{H})$.

□

3.3 The rationality criterion

We would like to state a theorem by Miyaoka that gives a complete numerical characterization of foliation by rational curves among foliations with non-dicritical singularities.

Theorem 3.3.1. *Let X be a smooth, complex algebraic surface and let $\mathcal{F}$ be a foliation on X. Suppose that there exists an ample divisor H on X such that*

$$T_{\mathcal{F}} \cdot H > 0.$$

Then, $\mathcal{F}$ is a foliation by rational curves, i.e. through each $x \in X$ there is a non-trivial rational curve tangent to $\mathcal{F}$; this means that up to blow-up, $\mathcal{F}$ is a rational fibration.

It is very important to stress that the converse of the theorem holds too: indeed, if if $\mathcal{F}$ is a foliation by rational curves, then there exists an ample H with $T_{\mathcal{F}} \cdot H > 0$. Let us construct such a divisor: let $\pi : X \to C$ be our foliation by rational curves, and let F be any regular fiber; thus, F is rational and the degree of K_F, its canonical bundle, is equal to -2. By adjunction formula, we have $deg K_F = (K_X + F) \cdot F = K_X \cdot F + F \cdot F$; since F is a fiber, we have $F \cdot F = 0$ and thus $K_X \cdot F = -2$. We know that $K_X = T^*_{\mathcal{F}} \otimes N^*_{\mathcal{F}}$, therefore $K_X \cdot F = T^*_{\mathcal{F}} \cdot F + N^*_{\mathcal{F}} \cdot F$. From a result of the previous chapter we have that, since F is a regular fiber and thus does not present any multiple components, $N^*_{\mathcal{F}} = \pi^* K_C$ and so $N^*_{\mathcal{F}} \cdot F = \pi^* K_C \cdot F = 0$. Thus, $T^*_{\mathcal{F}} \cdot F = -2$ and $T_{\mathcal{F}} \cdot F = 2 > 0$. F is not ample since $F \cdot F = 0$, but it is nef and therefore is on the border of the ample cone; hence, we can consider $H := nF + G$, with $n > 0$ and G ample, and we have that H is ample. Hence, $T_{\mathcal{F}} \cdot H = nT_{\mathcal{F}} \cdot F + T_{\mathcal{F}} \cdot G$, and taking n large enough this is certainly positive.

This criterion will be very useful in the next chapters and will be mainly used in the following form: if $\mathcal{F}$ is not a foliation by rational curves, then $T^*_{\mathcal{F}}$ is pseudoeffective, i.e. $T^*_{\mathcal{F}} \cdot H \geq 0$ for every ample H.

This result is particularly meaningful since it proves that the non-pseudoeffectiveness of $K_{\mathcal{F}}$ puts us in a very restricted class of foliations: we know that these foliations are algebraically integrable, meaning that they have algebraic leaves, and moreover these leaves are rational curves.

3.4 Numerical Kodaira dimension

Let us consider a foliation $\mathcal{F}$ on a smooth algebraic surface X and let us suppose that it is not a foliation by rational curves. Then due to Miyaoka's rationality

criterion, we know that $K_{\mathcal{F}}$ is pseudoeffective, i.e. $K_{\mathcal{F}} \cdot H \geq 0$ for every ample divisor H. A theorem of Fujita [11] proves that the canonical bundle $K_{\mathcal{F}}$ can be numerically decomposed as

$$K_{\mathcal{F}} \equiv P + N,$$

where

1. P is a nef $\mathbb{Q}$-divisor, i.e. $P = \sum_{j=1}^{p} a_j E_j$ with E_j irreducible curves, a_j rational numbers, and $P \cdot C \geq 0$ for every curve $C \subset X$;
2. N is an effective $\mathbb{Q}^+$-divisor, i.e. $N = \sum_{j=1}^{n} b_j D_j$, with D_j irreducible curves, b_j positive rational numbers, and the intersection matrix $(D_i \cdot D_j)$ negative definite, thus N is contractible;
3. $P \cdot D_j = 0$ for every $j = 1, ..., n$.

Recall that numerical equivalence $\equiv$ means that $K_{\mathcal{F}} \cdot C = P \cdot C + N \cdot C$ for every curve $C \subset X$. Recall that P and N are only $\mathbb{Q}$-divisors, not $\mathbb{Z}$-divisors, so in general they do not correspond to line bundles on X. The decomposition above is called the *Zariski decomposition of* $K_{\mathcal{F}}$, P is called *positive* or *nef* part, N is called *negative* or *contractible* part.

Definition 3.4.1. *We define the numerical Kodaira dimension $\nu(\mathcal{F})$ of $\mathcal{F}$ as*

- $\nu(\mathcal{F}) = 0$ *if* $P \equiv 0$*;*
- $\nu(\mathcal{F}) = 1$ *if* $P \not\equiv 0$*, but* $P^2 = 0$*;*
- $\nu(\mathcal{F}) = 2$ *if* $P^2 > 0$*;*
- $\nu(\mathcal{F}) = -\infty$ *if $K_{\mathcal{F}}$ is not pseudoeffective.*

Lemma 3.4.1. *Let $(X, \mathcal{F})$ be a foliation with reduced singularities on a smooth algebraic surface. Then, the numerical Kodaira dimension is a birational invariant.*

Proof. Let $\pi : X \dashrightarrow Y$ be a birational map between algebraic surfaces and let $\mathcal{F}$ be a foliation on X, $\mathcal{G}$ a foliation on Y. We need to prove that $\nu(\mathcal{F}) = \nu(\mathcal{G})$. Since every birational map is a composition of blow-ups and blow-downs, we can just assume that $\pi : \widetilde{X} \to X$ is a blow up.

Remark that if the canonical bundle $K_{\mathcal{F}}$ of a foliation $\mathcal{F}$ on X is not pseudoeffective, then there exists an ample divisor $H \in Pic(X)$ such that $H \cdot K_{\mathcal{F}} < 0$; consider the canonical bundle $K_{\widetilde{\mathcal{F}}}$ of the blown-up foliation $\widetilde{\mathcal{F}}$, then by projection formula we have $\pi^* H \cdot K_{\widetilde{\mathcal{F}}} = H \cdot \pi_* K_{\widetilde{\mathcal{F}}} = H \cdot K_{\mathcal{F}} < 0$, therefore $\nu(\mathcal{F}) = -\infty \Rightarrow \nu(\widetilde{\mathcal{F}}) = -\infty$. To prove the converse implication, we use again the projection formula: let $K \in Pic(\widetilde{X})$ such that $K \cdot K_{\widetilde{\mathcal{F}}} < 0$, then $\pi_* K \cdot K_{\mathcal{F}} = K \cdot \pi^* K_{\mathcal{F}} < 0$.

Suppose now that $\nu(\mathcal{F}) \geq 0$. Then, since the singularities of $\mathcal{F}$ are reduced, we have

$$K_{\widetilde{\mathcal{F}}} = \pi^* K_{\mathcal{F}} + E,$$

where E is an effective, π-exceptional divisor. From this and from the Zariski decomposition, we get

$$K_{\widetilde{\mathcal{F}}} \equiv \pi^* P + \pi^* N + E,$$

and we have that $\pi^* P$ is the positive part of the Zariski decomposition of $K_{\widetilde{\mathcal{F}}}$, while $\pi^* N + E$ is the negative part: in fact, $\pi^* P \cdot E = 0$ since E is π-exceptional, and if the matrix $(D_i \cdot D_j)_{ij}$ of the irreducible components of N is negative definite, then so is the matrix

$$\begin{pmatrix} (D_i \cdot D_j)_{ij} & 0 \\ 0 & E \cdot E \end{pmatrix}$$

since E is an exceptional divisor, so $E \cdot E < 0$. Therefore, $\pi^* P \cdot \pi^* P = P \cdot P$, and $P \equiv 0$, if and only if $\pi^* P \equiv 0$, which leads us to conclude that the numerical Kodaira dimension is invariant for blow-ups and thus is a birational invariant. □

Let us give an example of the computation of the numerical Kodaira dimension of the very special foliation $(Y, \mathcal{H})$. Let $(Y, \mathcal{H})$ be the very special foliation of Section 2.5.3, i.e. the resolution of a $\frac{\mathbb{Z}}{3\mathbb{Z}}$-quotient of a linear foliation $(\mathbb{C}P^2, \mathcal{L})$; recall that the resolution of each singular point $\hat{q}_j$ consists of two $\mathcal{H}$-invariant smooth curves, $D_j, E_j \subset Y$, each one of self-intersection -2, with $D_j \cdot E_j = 1$. Moreover, $Z(\mathcal{H}, D_j) = 1$ and $Z(\mathcal{H}, E_j) = 2$, i.e. $K_{\mathcal{H}} \cdot D_j = -1$ and $K_{\mathcal{H}} \cdot E_j = 0$. On a neighborhood of $D_j \cup E_j$ we have

$$\begin{cases} K_{\mathcal{H}} \cdot D_j = aD_j \cdot D_j + bE_j \cdot D_j = -2a + b \\ K_{\mathcal{H}} \cdot E_j = aD_j \cdot E_j + bE_j \cdot E_j = a - 2b. \end{cases}$$

Hence, $a = \frac{2}{3}$ and $b = \frac{1}{3}$ and $K_{\mathcal{H}} \equiv \frac{2}{3} D_j + \frac{1}{3} E_j$.

Since the foliation $\mathcal{L}$ has trivial canonical bundle, we have that $K_{\mathcal{H}}$ is numerically supported over $\bigcup_{j=1}^{3} D_j \cup E_j$: indeed,

$$K_{\mathcal{H}} \equiv \sum_{j=1}^{3} \left(\frac{2}{3} D_j + \frac{1}{3} E_j \right) = N$$

and in this case the positive part P is zero, so that $\nu(\mathcal{H}) = 0$.

3.4.1 The structure of the negative part

We have seen Hirzebruch-Jung strings before in this thesis; let us recall briefly what they are. We have a compact curve $C = \bigcup_{j=1}^{r} C_j$ where each C_j is a smooth rational curve of self-intersection ≤ -2, $C_i \cdot C_j = 1$ if $|i - j| = 1$, $C_i \cdot C_j = 0$ if $|i - j| \geq 2$.

Definition 3.4.2. *Let $(X, \mathcal{F})$ be a foliation; a curve $C \subset X$ is called an $\mathcal{F}$-chain if*

1. *C is an Hirzebruch-Jung string, $C = \bigcup_{j=1}^{r} C_j$;*

2. *each irreducible component C_j is $\mathcal{F}$-invariant;*

3. *$Sing(\mathcal{F}) \cap C$ are all reduced non-degenerate;*

4. *$Z(\mathcal{F}, C_1) = 1, Z(\mathcal{F}, C_j) = 2$ for every $j = 2, ..., r$.*

The most important result in this section is that, if $\mathcal{F}$ is relatively minimal, then the negative part of $K_{\mathcal{F}}$ is always supported on a disjoint union of $\mathcal{F}$-chains. Before we see the precise statement of the theorem though, we need two remarks:

1. Any $\mathcal{F}$-chain is contained in a maximal one, and two maximal $\mathcal{F}$-chains are necessarily disjoint: if this were not the case, their union would be a longer Hirzebruch-Jung string, except that $Z(\mathcal{F}, C_r) = 1$ instead of 2. But then, since a Hirzebruch-Jung string is contractible, if we contracted it to a point p we would obtain no separatrix through p, contradicting Camacho's Separatrix theorem ([3], section 3.3).

2. Any maximal $\mathcal{F}$-chain is contained in the support of the negative part N of $K_{\mathcal{F}}$: the first component C_1 because $K_{\mathcal{F}} \cdot C_1 = -1 < 0$, thus the second one also because $K_{\mathcal{F}} \cdot C_2 = 0$ and $C_1 \cdot C_2 > 0$, while $P \cdot C_1 = 0$, and so on. Let us observe that $K_{\mathcal{F}} \cdot C_j = 0$ for every $j = 2, ..., r$, by adjunction formula (Proposition 2.2.3): $K_{\mathcal{F}} \cdot C_j = -\chi(C_j) + Z(\mathcal{F}, C_j) = -2 + 2 = 0$.

Theorem 3.4.1. *Let $\mathcal{F}$ be a relatively minimal foliation on an algebraic surface X and suppose that $\mathcal{F}$ is not a foliation by rational curves. Let $K_{\mathcal{F}} \equiv P + N$ be the Zariski decomposition of its canonical bundle. Then, $Supp(N)$ is a disjoint union of maximal $\mathcal{F}$-chains.*

3.4.2 Foliations with vanishing numerical Kodaira dimension

We recall that a fibration $\pi : X \to B$ is said to be *isotrivial* if the general fibers of π are isomorphic.

In this paragraph, our aim is to present the beginning of a classification of those foliations that have numerical Kodaira dimension equal to zero. A very important result in this sense is the following:

Theorem 3.4.2. *Let $\mathcal{F}$ be a relatively minimal foliation on an algebraic surface X. Suppose that $\mathcal{F}$ is not a foliation by rational curves and that $\nu(\mathcal{F}) = 0$. Then, there exists a ramified covering $\pi : Y \to X$ and a birational morphism $p : Y \to Z$ such that $p_*\pi^*\mathcal{F}$ is generated by a global holomorphic vector field with isolated zeroes.*

We now present a result that is a consequence of the proof of the theorem, and will be very useful once we define the Kodaira dimension of a foliation:

Corollary 3.4.1. *Let $\mathcal{F}$ be a foliation on an algebraic surface X and suppose $\nu(\mathcal{F}) = 0$. Then, there exists $n \in \mathbb{N}$ such that $h^0(X, K_{\mathcal{F}}^{\otimes n}) > 0$.*

We saw in Section 2.6 which foliations are generated by global holomorphic vector fields; namely, these are

1. almost elliptic fiber bundles;
2. Kroenecker foliations on tori;
3. suspensions of representation $\rho : \pi_1(E) \to Aut(\mathbb{C}P^1)$, where E is an elliptic curve;
4. up to birational maps, foliations on $\mathbb{C}P^1 \times \mathbb{C}P^1$ generated by $v_1 \oplus v_2$, v_j holomorphic vector fields on $\mathbb{C}P^1$.

It would be interesting to obtain a list of all possible quotients of the foliations above; we need to study the automorphisms of these foliations, but we do not know if this would be sufficient, since we do not know if the ramified coverings appearing in the theorem are of Galois type.

3.4.3 Contraction of the negative part and canonical singularities

Our goal is to develop a minimal model program for the foliations similar to the one on surfaces, and thus we ask ourselves how can we make $K_{\mathcal{F}}$ as positive as possible, when possible: indeed, if our foliation is a foliation by rational curves, it will not be possible to obtain a nef model. The obstruction to this is clearly the negative part N in the Zariski decomposition of $K_{\mathcal{F}}$; if we were able to contract each component of its support to a point, then we would obtain a nef canonical bundle. However, the surface X_0 that we would get afterwards is, in general, not smooth anymore; it would have singularities of Hirzebruch-Jung type. The foliation $\mathcal{F}$ becomes a foliation $\mathcal{F}_0$, and we have $K_{\mathcal{F}_0}$ nef. The problem is that $K_{\mathcal{F}_0}$ is not a line bundle: on a neighborhood of p, the foliation $\mathcal{F}$ is not generated anymore by a holomorphic vector field with isolated zeroes. However, it can be proven that there exists an $m \in \mathbb{N}^+$ such that $K_{\mathcal{F}_0}^{\otimes m}$ is locally free at p (see [5], Section 1). In general, Hirzebruch-Jung singularities are very mild and the theory can be easily adapted to include them.

Definition 3.4.3. *Let $(X, \mathcal{F})$ be a foliation on a normal surface. A point $p \in X$ is a Hirzebruch-Jung singularity of $\mathcal{F}$ if p is a Hirzebruch-Jung singularity of X and on the minimal resolution $\pi : \widetilde{X} \to X$ the foliation $\mathcal{F}$ lifts to a foliation $\widetilde{\mathcal{F}}$ for which $\pi^{-1}(p)$ is an $\widetilde{\mathcal{F}}$-chain.*

Let us observe that, on a neighborhood of p, $\mathcal{F}$ is uniquely determined by the self-intersections of the Hirzebruch-Jung string resolving p.

We could now repeat all the theory developed in the last chapters in the slightly more general context of foliations on normal surfaces, all of whose singularities are either reduced or Hirzebruch-Jung singularities on singular points of the surface. The theory of minimal models can be adapted to this context, so we obtain the following result (see also [15]):

Theorem 3.4.3. *Any foliation on any normal algebraic surface, which is not birational to a Riccati foliation, to the very special foliation $\mathcal{H}$ or to a foliation by rational curves, is birational to a unique foliation $\mathcal{F}$ all of whose singularities are reduced or Hirzebruch-Jung, and whose canonical sheaf is nef.*

Such an $\mathcal{F}$ is constructed in the following way:

1. resolve all the singularities on the surface;
2. resolve the singularities of the foliation, using Seidenberg's theorem;
3. contract $\mathcal{F}$-exceptional curves;
4. contract maximal $\mathcal{F}$-chains.

In this situation, we are choosing the smallest class of singularities of the foliation in order to obtain a unique model with nef canonical sheaf; indeed, we need to ask for reduced singularities to ensure that the numerical Kodaira dimension and the Kodaira dimension (which will be defined in the next section) are birational invariants. We can also adopt another point of view, where we consider a larger class of singularities.

Definition 3.4.4. *Let $(X, \mathcal{F})$ be a foliation on a normal algebraic surface; $\mathcal{F}$ has canonical singularities if for any birational morphism $\pi : \hat{X} \to X$, we have*

$$K_{\hat{\mathcal{F}}} \equiv \pi^* K_{\mathcal{F}} + R,$$

where $\hat{\mathcal{F}} = \pi^*(\mathcal{F})$ and R is a $\mathbb{Q}^+$-divisor supported on the exceptional divisor of π.

If $p \notin Sing(X)$, then if p is either regular or reduced for the foliation, it is certainly canonical: indeed, when we blow up we obtain $K_{\hat{\mathcal{F}}} = \pi^* K_{\mathcal{F}} + aE$, where $a = 1$ if p is regular and $a = 0$ if p is reduced singular. The converse is not exactly true: for example, the Poincaré-Dulac foliation given by the holomorphic 1-form $(nw + z^n)dz - zdw = 0$ does not have a reduced singularity in $p = (0, 0)$, but it is canonical: indeed, after n blow-ups we either get a reduced singularity, or

$$\pi_0^{n*}\omega = z[dz - zdw],$$

therefore $l(p) = 1$ and $K_{\hat{\mathcal{F}}} = \pi^* K_{\mathcal{F}}$.

The case $p \in Sing(X)$ is much more complicated: indeed, Hirzebruch-Jung singularities are canonical, but they are not the only one. However, in this thesis we only work with either smooth surfaces or surfaces which present Hirzebruch-Jung singularities; for a detailed description of canonical singularities, see [15].

3.5 Kodaira dimension

Given a line bundle L on an algebraic surface X, its Kodaira dimension is defined as

$$\mathrm{kod}(L) = \begin{cases} -\infty & \text{if } h^0(X, L^{\otimes n}) = 0 \ \forall n \in \mathbb{N}^+ \\ \max_{d \in \mathbb{N}} \dim \varphi_{|dL|}(X) & \text{otherwise.} \end{cases}$$

In other words, $\text{kod}(L) \geq k$ if and only if for some $n \in \mathbb{N}^+$ there are $k+1$ non-trivial sections $s_j \in H^0(X, L^{\otimes n}), j = 0, ..., k$ which are algebraically independent. This is equivalent to saying that we have a dominant rational map

$$\begin{aligned} X &\longrightarrow \mathbb{C}P^k \\ x &\mapsto (s_0(x) : ... : s_k(x)). \end{aligned}$$

Since L is a line bundle on an algebraic surface, we have $\text{kod}(L) \in \{-\infty, 0, 1, 2\}$. If L is pseudoeffective, we can also consider its numerical Kodaira dimension $\nu(L)$ and, as we have done above, we set $\nu(L) = -\infty$ if L is not pseudoeffective. We will prove later that

$$\text{kod}(L) \leq \nu(L);$$

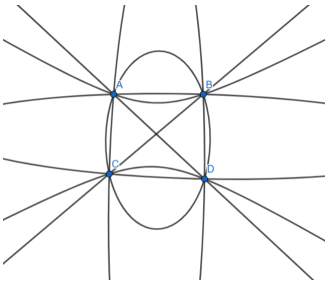
if $\text{kod}(L) = \nu(L)$, we say that there is *abundance*.

Now, let $(X, \mathcal{F})$ be a foliation on X.

Definition 3.5.1. *The Kodaira dimension* $\text{kod}(\mathcal{F})$ *of* $\mathcal{F}$ *is the Kodaira dimension of its canonical bundle,* $\text{kod}(K_{\mathcal{F}})$.

Proposition 3.5.1. *Let* $(X, \mathcal{F})$ *be a foliation with reduced singularities. Then,* $\text{kod}(\mathcal{F})$ *is a birational invariant.*

Let us see an example of what happens if the singularities are not reduced: let us take $\mathcal{F}$ to be a pencil of conics in $\mathbb{C}P^2$, so we take four points and all the possible conics through all of them:



since $\mathcal{F}$ is not a pencil of lines, we have $K_{\mathcal{F}} = \mathcal{O}_{\mathbb{C}P^2}(2)$, so its Kodaira dimension is non-negative; moreover, the singularities are dicritical and thus non-reduced, thanks to [18], Theorem 8. If we resolve the singularities, we obtain four exceptional divisors. i.e. four lines, that separate all the conics: this is now a foliation by rational curves, so by Miyaoka's theorem, its line bundle $K_{\widetilde{\mathcal{F}}}$ is not pseudoeffective and thus, its Kodaira dimension is $-\infty$.

We need to remark two particular cases of abundance:

1. If $\nu(\mathcal{F}) = 2$, then $\text{kod}(\mathcal{F}) = 2$; this is a general property that holds for any pseudoeffective line bundle L, and it comes from Riemann-Roch formula. Indeed, suppose that L is a pseudoeffective line bundle and consider its

Zariski decomposition $L \equiv P + N$. Take $m \in \mathbb{N}$ such that mP is Cartier; then, from Riemann-Roch formula we get

$$h^0(X, mP) - h^1(X, mP) + h^2(X, mP) = \chi(mP) = \chi(\mathcal{O}) + \frac{1}{2}mP(-K_X + mP) \tag{3}$$

and thus

$$\frac{m^2P^2}{2} - \frac{mKP}{2} + \chi(\mathcal{O}) \longrightarrow +\infty \text{ for } m \longrightarrow +\infty.$$

This means that we can choose a sufficiently large m such that $h^0(X, mP) + h^2(X, mP) > 0$. Our aim is to prove that $h^0(mP)$ grows like m^2, i.e. $h^0(mP) = \mathcal{O}(m^2)$; from the expressions above, it suffices to show that $h^2(mP) = \mathcal{O}(m)$. From Serre duality, we have $h^2(mP) = h^0(K_X - mP)$. Now, let C be a curve such that its class in the Néron-Severi group is ample; then, $P \cdot C > 0$ and moreover, if $D \in NS(X)$ is such that $D \cdot C < 0$, then D is not an effective divisor. Let us observe that, for m sufficiently big, $(K_X - mP) \cdot C = K_X \cdot C - mP \cdot C < 0$, thus $K_X - mP$ is not effective and does not have any sections.

Observe that from (3) we get that the converse is also true: indeed, if $\mathrm{kod}(L) = 2$, we have that $h^0(X, mP)$ grows like a quadratic polynomial in m, but if $\nu(L) < 2$ then $P \cdot P = 0$ and this is absurd because on the right hand side of the equation we would have a polynomial that grows linearly in m. Therefore,

$$\mathrm{kod}(L) = 2 \Longleftrightarrow \nu(L) = 2.$$

2. $\nu(\mathcal{F}) = 0$ if and only if $\mathrm{kod}(\mathcal{F}) = 0$: we will see this in more details later.

From this previous analysis, it is evident that the only non-abundant case can happen when $\nu(\mathcal{F}) = 1$; we will see later on what can be said about this particular situation.

We can finally prove the following

Lemma 3.5.1. *Let X be an algebraic surface and let L be a line bundle on X. Then,*

$$\mathrm{kod}(L) \leq \nu(L).$$

Proof. We divide the proof in four cases:

- In case $\mathrm{kod}(L) = -\infty$, we have nothing to prove.
- In case $\mathrm{kod}(L) = 0$, then $\nu(L) \geq 0$: indeed, if it were $\nu(L) = -\infty$, then by definition L would not be pseudoeffective, but then L would also not be effective and thus the Kodaira dimension would be $-\infty$.
- We have proved already that $\mathrm{kod}(L) = 2$ if and only if $\nu(L) = 2$.

- By the previous case, if $\mathrm{kod}(L) = 1$, then $\nu(L) \leq 1$; we need to prove that $P \not\equiv 0$, where P is the positive part of the Zariski decomposition. From the fact that $\mathrm{kod}(L) = 1$ we know that there exists an $m \in \mathbb{N}$ such that $h^0(X, mL) = h^0(X, mP) = 2$, meaning that there exist two effective and linearly independent divisors D_1, D_2 that are linearly equivalent to mP. Let H be the class of an ample divisor on X, then from Nakai-Moishezon criterion we get $D_1 \cdot H = mP \cdot H > 0$, and thus we have that P is not numerically zero.

□

3.5.1 Foliations of Kodaira dimension 1

In this section, our aim is to give a precise characterization of foliations with Kodaira dimension 1.

Theorem 3.5.1. *[3, Theorem 9.1] Let $\mathcal{F}$ be a foliation with reduced singularities on an algebraic surface X such that $\mathrm{kod}(\mathcal{F}) = 1$. Then, $\mathcal{F}$ belongs to one of the following classes:*

1. *Riccati foliations;*
2. *turbulent foliations;*
3. *non-isotrivial elliptic fibrations;*
4. *isotrivial fibrations of genus ≥ 2.*

The idea of the proof is to construct the Iitaka fibration, and then observe that $\mathcal{F}$ can either coincide with it or be generally transverse to it; in the former case, we have *3.*, in the latter one we obtain *1.,2.,4.,* depending on the genus of a generic fibre F of the fibration.

Conversely, all the foliations above mentioned have Kodaira dimension ≤ 1; it can happen that a Riccati or a turbolent foliation has Kodaira dimension equal to zero or $-\infty$, but these cases are quite exceptional.

3.5.2 Foliations of Kodaira dimension 0

In this chapter, we completely characterize foliations of Kodaira dimension 0 as foliations of numerical Kodaira dimension 0; in other words, we state that

$$\nu(\mathcal{F}) = 0 \iff \mathrm{kod}(\mathcal{F}) = 0.$$

We have seen before the "if" implication; the next theorem proves the "only if" implication. A proof of this result can be found in [3, Theorem 9.2].

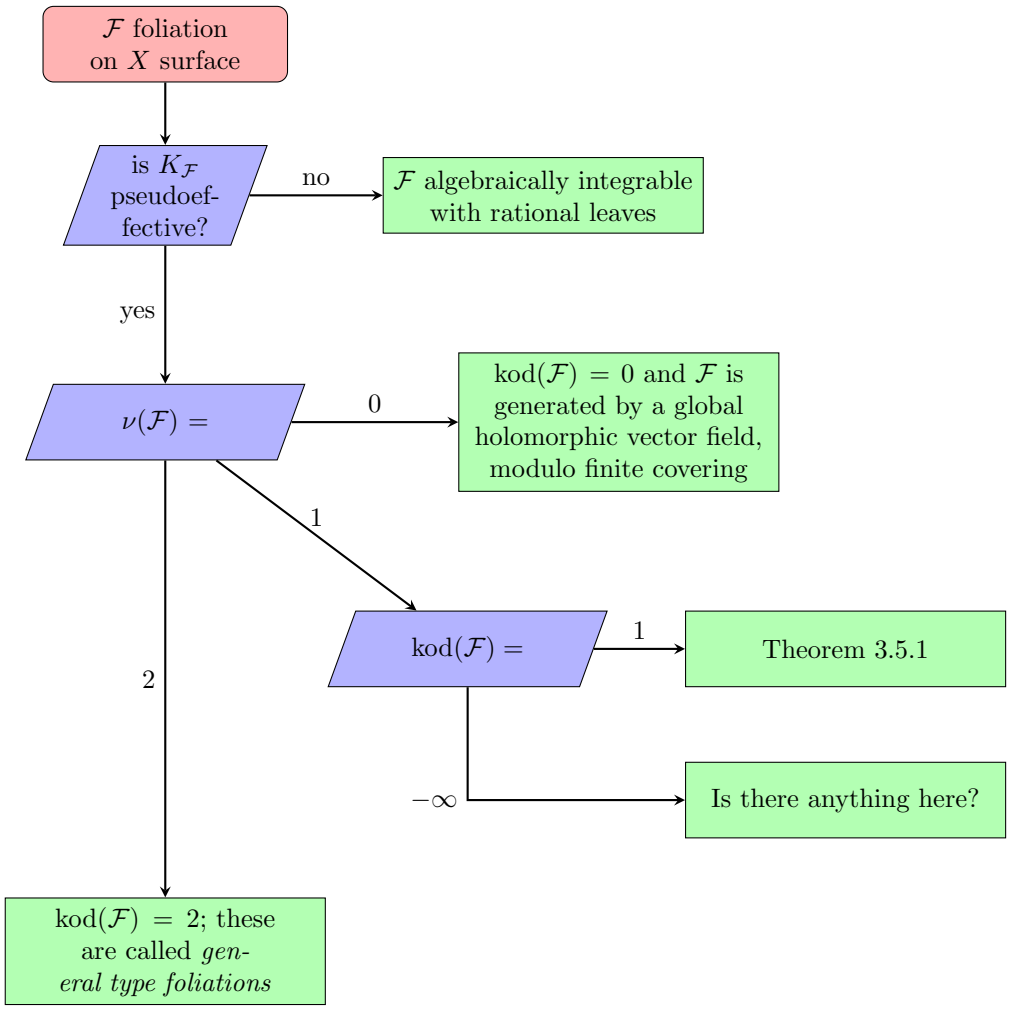
Theorem 3.5.2. *Let $\mathcal{F}$ be a foliation with reduced singularities on an algebraic surface X, and suppose $\mathrm{kod}(\mathcal{F}) = 0$. Then, $\nu(\mathcal{F}) = 0$.*

The most important corollary of this is the following one:

Corollary 3.5.1. *Let $\mathcal{F}$ be a foliation with reduced singularities on a complex algebraic surface X, and suppose* $\mathrm{kod}(\mathcal{F}) = 0$. *Then, up to a ramified covering and up to birational maps, $\mathcal{F}$ is generated by a global holomorphic vector field with isolated zeros.*

3.6 Summary

Let us explain in a graph what the classification of foliations on a complex algebraic surface X looks like so far.



We are going to answer the question "is there anything here?" in the next section.

4 Non-abundant foliations

The aim of this section is to study foliations $\mathcal{F}$ such that $\mathrm{kod}(\mathcal{F}) < \nu(\mathcal{F})$. Firstly, we will give an example of a class of those, the *Hilbert modular foliations*; after that, we will prove that these are the only non-abundant foliations. To this end, we will present a program given by Marco Brunella at the end of his book [3]; then, to complete the proof, we will analyze the article [5] by Marco Brunella, in which he defines the Poincaré metric on each leaf of the foliation by looking at its universal covering. He then uses that to construct a singular hermitian metric on the canonical bundle of the foliation, the so-called *canonical Poincaré metric*. In the Main Theorem, it is shown that the curvature of this metric is a closed positive $(1,1)$-current Ω, and thus we can use this framework to construct a foliation $\mathcal{G}$ transverse to a non-abundant foliation $\mathcal{F}$ lying in the kernel of Ω. Finally, we will prove that this foliation $\mathcal{G}$ is holomorphic, thereby proving that $\mathcal{F}$ is a Hilbert modular foliation.

4.1 Hilbert modular foliations

The only case that we have not analyzed yet in our classification is the non-abundant one. We shall see that the class of foliations with numerical Kodaira dimension equal to 1 and negative Kodaira dimension is not empty. Let us introduce two examples:

Example 4.1.1. Let X be a compact surface whose universal cover is the bidisc $\mathbb{D}\times\mathbb{D}$. The automorphism group of $\mathbb{D}\times\mathbb{D}$ preserves the vertical and the horizontal foliations, so that X naturally has two (non-singular) fibrations $\mathcal{F}$ and $\mathcal{G}$, one transverse to the other. If $\pi_1(X) = \Gamma \subset Aut(\mathbb{D}\times\mathbb{D})$ is not, up to finite index, a product $\Gamma_1\times\Gamma_2$, $\Gamma_i \subset \mathrm{Aut}(\mathbb{D})$, then both $\mathcal{F}$ and $\mathcal{G}$ have dense leaves in X.

In that case, we have $\nu(\mathcal{F}) = \nu(\mathcal{G}) = 1$ but $\mathrm{kod}(\mathcal{F}) = \mathrm{kod}(\mathcal{G}) = -\infty$. The computation of ν follows from the fact that $\mathcal{F}$ is non-singular and thus relatively minimal; we have proved before that if we have a relatively minimal foliation, then the negative part of $K_{\mathcal{F}}$ is supported on a disjoint union of maximal $\mathcal{F}$-chains. But since the foliation is non-singular, we do not have any $\mathcal{F}$-chain on our surface: $Z(\mathcal{F}, C) = 0$ for any $\mathcal{F}$-invariant curve C. Therefore, N is empty, so $K_{\mathcal{F}} \equiv P$. We have an isomorphism $K_{\mathcal{F}} \simeq N^*_{\mathcal{G}}$ due to the fact that the two foliations are mutually transverse, and $\mathcal{G}$ is a regular foliation, therefore applying Baum-Bott formula (see [3], chapter 3) to $N^*_{\mathcal{G}}$ we get

$$0 = N^*_{\mathcal{G}} \cdot N^*_{\mathcal{G}} = K_{\mathcal{F}} \cdot K_{\mathcal{F}} = P \cdot P;$$

moreover, from [3], Proposition 2.1 we deduce $P \cdot K_X = c_2(X)$, and the latter is positive due to [10], Theorem 5.21 and Example 5.23; this leads us to the conclusion that $\nu(\mathcal{F}) = 1$.

The computation of the Kodaira dimension follows from the isomorphism $K_{\mathcal{F}} \simeq N^*_{\mathcal{G}}$. Bogomolov's Lemma (see Proposition 2.6.1) excludes $\mathrm{kod}(\mathcal{F}) \geq 1$, and we cannot have $\mathrm{kod}(\mathcal{F}) = 0$ by Section 3.5.2, since in particular this would

mean $\nu = 0$. Note also that if Γ is a product $\Gamma_1 \times \Gamma_2$ (up to finite index) then $\mathcal{F}$ and $\mathcal{G}$ are both isotrivial fibrations of genus ≥ 2, since their leaves have the disk $\mathbb{D}$ as universal covering.

Example 4.1.2. Let X be a Hilbert modular surface: it is a smooth compactification of an open surface $\widetilde{X}_0$ which is the (minimal) resolution of $X_0 = \mathbb{D}\times\mathbb{D}/\Gamma$, where $\Gamma \subset Aut(\mathbb{D}\times\mathbb{D})$ is a special subgroup, which we ask to be cocompact, torsion free and irreducible. Singularities of X_0 are quotient singularities, resolved by Hirzebruch-Jung strings. The compactification of $\widetilde{X}_0$ is done by adding cycles of rational curves, as in [15], Example II.2.3. As in the previous example, we have on X two natural foliations $\mathcal{F}$ and $\mathcal{G}$, originated by the two fibrations of $\mathbb{D} \times \mathbb{D}$.

These foliations have some compact leaves: the Hirzebruch–Jung strings, which are moreover maximal $\mathcal{F}$-chains and $\mathcal{G}$-chains, and the cycles of rational curves, on which $\mathcal{F}$ and $\mathcal{G}$ are singular only at crossing points, the singularities being reduced and non-degenerate. Outside these compact leaves, $\mathcal{F}$ and $\mathcal{G}$ are non-singular and transverse to each other. Hence we have

$$K_{\mathcal{F}} \otimes N_{\mathcal{G}} = \mathcal{O}_X(D), \tag{4}$$

where D is a positive integral divisor (the so-called tangency divisor between $\mathcal{F}$ and $\mathcal{G}$, locally defined by $\omega(v) = 0$, being ω a local 1-form generating $\mathcal{G}$ and v a local vector field generating $\mathcal{F}$) whose support is equal to the union of the compact leaves since those are both $\mathcal{F}$ and $\mathcal{G}$ invariant and the foliations are transverse outside of them. If P is the positive part of $K_{\mathcal{F}}$ from (4) above and from the fact that N is supported on Hirzebruch-Jung strings we also see that $P \simeq N_{\mathcal{G}}^* \otimes \mathcal{O}_X^*(Z)$, where Z is the reduced union of the cycles of rational curves. We can apply the same arguments as in the previous example, in their logarithmic version, to conclude that $\mathrm{kod}(\mathcal{F}) = -\infty$ and $\nu(\mathcal{F}) = 1$, and we can do the same for $\mathcal{G}$.

These two classes of examples are both called *Hilbert modular foliations*.

Thus, we have seen that the answer to the question "do non-abundant foliations exist?" is positive, and this is a striking difference between the classification of surfaces and foliations on surfaces: indeed, in the case of foliations we have that the canonical bundle $K_{\mathcal{F}}$ can be nef, but not effective, while this does not happen for surfaces, where the nefness of the canonical bundle K_X automatically implies its semi-ampleness.

The next question that we want to answer is:

Question

Is there any non-abundant foliation that is not a Hilbert modular foliation?

The answer is no; the proof of this was obtained by Brunella and McQuillan after Brunella's book [3] was released. In the remaining part of this section, we will see how Brunella approached the proof of this problem in [3]; the next

chapter will be dedicated to the presentation of the complete proof that can be found in [5].

Let $\mathcal{F}$ be a relatively minimal foliation on an algebraic surface X such that $\mathrm{kod}(\mathcal{F}) = -\infty$ and $\nu(\mathcal{F}) = 1$. We have that $h^1(X, \mathcal{O}_X) = 0$: otherwise, from Remark 2.6.1 we would have that $\mathcal{F}$ is the Albanese fibration, a contradiction. In order to prove that $\mathcal{F}$ is a Hilbert modular foliation, we want to find its companion $\mathcal{G}$. Up to the resolution of eventual singularities of the underlying surface X, we can suppose that $K_{\mathcal{F}}$ is nef, thus $K_{\mathcal{F}} \equiv P$ and in particular this means $K_{\mathcal{F}} \cdot K_{\mathcal{F}} = 0$. A result of Demailly ([8]) proves that there exists on $K_{\mathcal{F}}$ a singular hermitian metric whose curvature Ω is a closed positive current; this current numerically represents $K_{\mathcal{F}}$, but it cannot be the integration current over a $\mathbb{Q}^+$-divisor, because that would mean that the Kodaira dimension of $\mathcal{F}$ is non-negative. But it can be proved that the Lelong numbers of Ω are zero everywhere. Write $[\Omega]$ for the cohomology class represented by Ω, then the fact that the Lelong numbers are zero everywhere and that $[\Omega]^2 = 0$, suggests that Ω defines through its kernel a foliation $\mathcal{G}$ on X, called *Monge-Ampère foliation*: if we add some regularity hypothesis, then the condition $[\Omega]^2 = 0$ translates into $\Omega \wedge \Omega = 0$, so the kernel of the (1,1)-form Ω_x is a one-dimensional complex line $L_x \subset T_x X$, and then this distribution of lines is integrable due to the closedness of Ω. But there are many obstacles in this construction:

- we do not know if our current is sufficiently regular to define the wedge product $\Omega \wedge \Omega$;
- even if Ω is sufficiently regular, the foliation $\mathcal{G}$ is defined only where Ω is not 0, and we do not know if this set is dense in X;
- the foliation $\mathcal{G}$ has complex leaves, but it might not be holomorphic.

If we suppose that $\mathcal{G}$ exists and it is holomorphic, or, more precisely, we suppose that there exists on X a holomorphic foliation $\mathcal{G}$ such that the closed positive current Ω is $\mathcal{G}$-invariant, then we can prove that $\mathcal{F}$ and $\mathcal{G}$ are the Hilbert modular foliations that we were looking for: indeed, Ω is diffuse and $[\Omega]^2 = 0$, hence from Theorem 2.3.1 we deduce that $N_{\mathcal{G}} \cdot [\Omega] = 0$, i.e. $N_{\mathcal{G}} \cdot K_{\mathcal{F}} = 0$. On the other hand, we have $N_{\mathcal{F}} \cdot K_{\mathcal{F}} = -K_X \cdot K_{\mathcal{F}} < 0$, otherwise by Riemann-Roch formula and by the fact that $h^1(X, \mathcal{O}_X) = 0$ we would reach a contradiction regarding the Kodaira dimension of the foliation. This means, in particular, that $\mathcal{G}$ is different from $\mathcal{F}$; we can thus define their tangency divisor D, which satisfies

$$\mathcal{O}_X^*(D) = K_{\mathcal{F}} \otimes N_{\mathcal{G}}.$$

We have $K_{\mathcal{F}} \cdot K_{\mathcal{F}} = 0$ and $K_{\mathcal{F}} \cdot N_{\mathcal{G}} = 0$, thus $K_{\mathcal{F}} \cdot D = 0$; let $D = \sum_{j=1}^m D_j$ be the decomposition of D into irreducible curves, then we have, since $K_{\mathcal{F}}$ is nef, that $K_{\mathcal{F}} \cdot D_j = 0$ for every j. From this relation and Hodge index theorem we have $D_j^2 < 0$ for every j, and therefore each D_j is $\mathcal{F}$-invariant: otherwise, we would have $K_{\mathcal{F}} \cdot D_j = -D_j^2 + tang(\mathcal{F}, D_j) > 0$. This forces D_j to be either a smooth rational curve with $Z(\mathcal{F}, D_j) = 2$, or a rational curve with a node and $Z(\mathcal{F}, D_j) = 0$.

Since D_j are the curves on which $\mathcal{F}$ and $\mathcal{G}$ are not transverse, the fact that they are $\mathcal{F}$-invariant means that they also are $\mathcal{G}$-invariant. Hence, $N_{\mathcal{G}} \cdot D_j = D_j^2 + Z(\mathcal{G}, D_j)$, i.e. $N_{\mathcal{G}} \otimes \mathcal{O}_X^*(-D_{\text{red}}) \cdot D_j = Z(\mathcal{G}, D_j) - (\bigcup_{k \neq j} D_k) \cdot D_j$. Each intersection of D_j with D_k, with $k \neq j$, gives a positive contribution to $Z(\mathcal{G}, D_j)$, so

$$N_{\mathcal{G}} \otimes \mathcal{O}_X^*(-D_{\text{red}}) \cdot D_j \geq 0$$

for all j.

Since $K_{\mathcal{F}}$ is numerically trivial on a neighborhood of the support of D, we have $N_{\mathcal{G}} \equiv \mathcal{O}_X^*(D)$ on the same neighborhood, and so $N_{\mathcal{G}} \otimes \mathcal{O}_X^*(-D_{\text{red}})$ is numerically non-negative. This, the inequalities above and the negativity of the intersection form on D_js means that $N_{\mathcal{G}} \otimes \mathcal{O}_X^*(-D_{\text{red}})$ is actually numerically trivial, or equivalently $D = D_{\text{red}}$. Thus we have the equality $Z(\mathcal{G}, D_j) = (\bigcup_{k \neq j} D_k) \cdot D_j$, i.e $Sing(\mathcal{G}) \cap D = Sing(D)$, all non-degenerate.

A connected component of D is a chain or a cycle of rational curves, but it cannot be a chain. If that were the case, we can proceed like in the previous chapter to reach a contradiction: we would have $Z(\mathcal{G}, D_j) = 2$ for every j in the chain; taking a singularity p on the handle of the chain, if we wished to contract the chain, then the image of p under the contraction would have no separatrix through it, contradicting Camacho's version of the separatrix theorem.

Therefore, each connected component of D is a cycle of rational curves, $Sing(\mathcal{F}) \cap D = Sing(\mathcal{G}) \cap D = Sing(D)$ and these are all reduced. $\mathcal{F}$ and $\mathcal{G}$ are transverse to each other outside of D and tangent at first order along D. We want to prove that $\mathcal{F}$ and $\mathcal{G}$ are the two canonical foliations on a Hilbert modular surface; the above properties of the tangency divisor between $\mathcal{F}$ and $\mathcal{G}$ mean that $\Omega_X^1(\log(D))$ splits as a direct sum $K_{\mathcal{F}} \oplus K_{\mathcal{G}}$ and moreover, $\det \Omega_X^1(\log(D)) = K_{\mathcal{F}} \otimes K_{\mathcal{G}} = L$ is nef and big, i.e. $L^2 > 0$ (this follows from $K_{\mathcal{G}} \cdot K_{\mathcal{F}} = (K_X + N_{\mathcal{G}}) \cdot K_{\mathcal{F}} > 0$). Suppose that C is a curve not contained in D; then, it is not $\mathcal{F}$-invariant, so $K_{\mathcal{F}} \cdot C > 0$ and $L \cdot C > 0$. By a result of Tian and Yau in [19], X is a Hilbert modular surface and $\mathcal{F}, \mathcal{G}$ are Hilbert modular foliations.

4.2 The leaf-wise Poincaré metric

From now on, our main reference for all the theorems we present will be [5], unless stated otherwise. Let X be a compact complex algebraic surface, and let $\mathcal{F}$ be a holomorphic foliation, possibly with singularities, on X. Unlike before, we do not assume that X is smooth; instead, we are going to allow for *Hirzebruch-Jung singularities*, and this assumption will enable us to contract the negative part of the Zariski decomposition of $K_{\mathcal{F}}$ (provided that the foliation $\mathcal{F}$ is not a rational fibration). We do not impose conditions on the singularities of $\mathcal{F}$, other than they do not intersect the singularities of X. We define

$$X' = X \setminus \text{Sing}(\mathcal{F}) \qquad X'' = X' \setminus \text{Sing}(X)$$

Our goal is to provide a precise description of a generic leaf of the foliation $\mathcal{F}$. Before discussing Brunella's construction, we adopt the following conventions in regards to the use of the word *leaf*:

- If $p \in X''$, then the definition of local leaf through p is the usual one.
- If $p \in \mathrm{Sing}(X)$, then, locally, X looks like the quotient of a ball $\mathbb{B}^2$ by a finite cyclic group $\mathbb{Z}_n$, where the $\mathbb{Z}_n$-action on $\mathbb{B}^2$ is generated by $(z,w) \mapsto (e^{2\pi i \frac{q}{n}} z, e^{2\pi i \frac{1}{n}} w)$ for some $q = 1, ..., n-1$, coprime with n, and $\mathcal{F}$ is the quotient of the vertical foliation $\{dz = 0\}$ on $\mathbb{B}^2$. Thus, we can define the local leaf of $\mathcal{F}$ through p as the projection of $\{z = 0\}$ under $\mathbb{B}^2 \to \mathbb{B}^2/\mathbb{Z}_n$. Hence, this local leaf is a disc centered at p, but it must be considered as an orbifold, with p affected by the multiplicity n.
- If $p \in \mathrm{Sing}(\mathcal{F})$, then the local leaf through p is, by definition, the single point $\{p\}$.

Thus, we can see that for every $p \in X$ there exists a well-defined leaf of $\mathcal{F}$ through p, which is $\{p\}$ if $p \in \mathrm{Sing}(\mathcal{F})$ and a connected orbifold L_p containing p and immersed in X' if $p \in X'$. Even if $p \in Sing(X)$, the orbifold L_p is not a bad orbifold, i.e. it always admits a universal covering, since it is always the quotient of $\mathbb{B}^2$ by the properly discontinuous action of $\mathbb{Z}_n$, and thus if $p \in X'$, we can see that L_p always admits a universal covering $\widetilde{L}_p$. Thus, if $p \in X'$, we can define on L_p its Poincaré metric:

- if $\widetilde{L}_p \simeq \mathbb{D}$, then L_p carries the unique complete metric of constant curvature -1;
- if $\widetilde{L}_p \simeq \mathbb{C}$ or $\mathbb{C}P^1$, then the Poincaré metric is identically zero.

Let now us fix $p \in X''$ and local coordinates (z,w) centered at p such that $\mathcal{F} = \{dz = 0\}$. Then, the volume form associated to the area of the leaf-wise Poincaré metric is locally expressed as

$$e^{F(z,w)} i dw \wedge d\overline{w} \tag{5}$$

for some function F with values in $[-\infty, +\infty)$.

Consider a leaf $\{z = z_0\}$: if it is a parabolic leaf, then F is identically $-\infty$. Otherwise, by definition of Poincaré metric, we must impose the condition that the curvature is -1; recall that, in complex dimension 1, the Gaussian curvature K of an Hermitian metric

$$ds^2 = \lambda(z)^2 |dw|^2$$

is given by

$$K(z) = -\frac{4}{\lambda(z)^2} \frac{\partial^2}{\partial w \partial \overline{w}} \log \lambda(z),$$

thus

$$-1 = -\frac{1}{e^F} \frac{\partial^2 F}{\partial w \partial \overline{w}},$$

thus F satisfies the elliptic partial differential equation

$$\frac{\partial^2 F}{\partial w \partial \overline{w}} = e^F \tag{6}$$

and thus it is real analytic, since solutions of elliptic PDEs are analytic (see [17]).

From this local analysis, we see that $F(z_0, \cdot)$ is either subharmonic or identically $-\infty$ for every fixed z_0. The goal of the main theorem is to prove that F is plurisubharmonic. We can prove the upper-semicontinuity of F by looking at the definition of the metric on the leaves as the Kobayashi metric, which coincides with the Poincaré metric for Riemann surfaces. Let $p \in L$ a point on a leaf, and $v \in T_pL$; then, we define the Kobayashi metric

$$\rho_L(p; v) = \inf \left\{ \lambda > 0 | \exists f : \mathbb{D} \longrightarrow L, f(0) = p, f'(0) = \frac{v}{\lambda} \right\}.$$

If we have a parabolic leaf, then the Kobayashi metric is identically 0, whereas if the leaf is hyperbolic, it has constant curvature -1, thus it coincides with the Poincaré metric (see [12, Chapter 3]); then, from (6) we get

$$\left|\left| \frac{\partial}{\partial w} \right|\right|^2_{Kob} = e^F(z, w),$$

thus

$$F(z, w) = \log \left(\rho_{L_z} \left((z, w); \frac{\partial}{\partial w} \right)^2 \right)$$

and F is upper semi-continuous if and only if the Kobayashi norm is; however, this follows from the definition since $\rho_L(p; v)$ is defined as an infimum.

Concerning $K_{\mathcal{F}}$, we must remark that it is not necessarily locally free around a point in $\mathrm{Sing}(X)$; however, it is always an invertible sheaf associated to a $\mathbb{Q}$-Cartier Weil divisor. If $K_{\mathcal{F}}$ is nef, we say that $\mathcal{F}$ is a *nef foliation*. Let us observe that, in this case, every leaf L of $\mathcal{F}$ has universal covering isomorphic to $\mathbb{D}$ or to $\mathbb{C}$: indeed, if $\widetilde{L} \simeq \mathbb{C}P^1$, then L is an algebraic curve of genus 0, and thus

$$K_{\mathcal{F}} \cdot L = -\chi(L) + Z(\mathcal{F}, L) = -2 + Z(\mathcal{F}, L) < 0$$

by adjunction formula (Proposition 2.2.3) which contradicts the nefness of $K_{\mathcal{F}}$.

We want to look at the leaf-wise Poincaré metric as a singular hermitian metric on $K_{\mathcal{F}}$. Around $p \in X''$, we may choose a local non-vanishing holomorphic vector field $v = \frac{\partial}{\partial w}$ generating $\mathcal{F}$; then, if we compute the squared norm of v with respect to the leaf-wise Poincaré metric, we obtain

$$||v||^2 = e^F dw(v)\overline{dw(v)} = e^F,$$

where F takes values in $[-\infty, +\infty)$ and satisfies

$$\mathcal{L}_v \mathcal{L}_{\overline{v}} F = \mathcal{L}_{\frac{\partial}{\partial w}} \mathcal{L}_{\frac{\partial}{\partial \overline{w}}} = \frac{\partial^2 F}{\partial w \partial \overline{w}} = e^F,$$

where $\mathcal{L}_v, \mathcal{L}_{\overline{v}}$ are the Lie derivatives.

If $p \in \mathrm{Sing}(\mathcal{F})$, then we can take a local vector field vanishing only at p and generating $T_{\mathcal{F}}$, and the same holds, provided that we can define F at p.

If $p \in Sing(X)$, we need to replace $T_{\mathcal{F}}$ with $T_{\mathcal{F}}^{\otimes n}$ to ensure local freeness, but then we have no problem defining F and the same holds. Thus, the function F is the local weight of a singular hermitian metric on $T_{\mathcal{F}}\big|_{X'}$, and by dualizing we obtain a singular hermitian metric on $K_{\mathcal{F}}\big|_{X'}$, which will be called the *canonical Poincaré metric*. Its curvature Ω is given by

$$\Omega = \frac{i}{2\pi}\partial\overline{\partial}F,$$

and if the local weights F are locally integrable, then Ω is a well-defined closed (1,1)-current on X', which represents the first Chern class of $K_{\mathcal{F}}$. We say that the curvature is *positive* if Ω is a closed positive current, i.e. the local weights F are plurisubharmonic; from the definition, we have that F is upper semi-continuous. This is of course true if $F \equiv -\infty$, so suppose we are on a hyperbolic leaf. Then, if we take all the holomorphic functions $f : \mathbb{D} \to L_{z_0}$ such that $0 \mapsto w$, where L_{z_0} is the leaf defined by $\{z = z_0\}$, we have that

$$e^{F(z,w)} = \sup_f |df(0))|^2$$

and thus

$$F(z,w) = 2log \sup_f |df(0)| = 2 \sup_f log|df(0)|,$$

since the logarithm is a continuous and monotone increasing function, and thus we can exchange the supremum and the logarithm. But now F is the supremum of a continuous function, thus is an upper semi-continuous function.

Let us observe that, according to the extension results in [6], cap. 5.F, we have that if Ω is positive, then it extends to the full X as a closed positive current, and thus the canonical Poincaré metric is globally defined.

4.3 Main theorem

We are now ready to state our main result.

Theorem 4.3.1 (Main Theorem)**.** *Let $\mathcal{F}$ be a nef foliation on a compact complex algebraic surface X and suppose that there exists at least one hyperbolic leaf. Then, the canonical Poincaré metric has positive curvature.*

We require the existence of at least one hyperbolic leaf because otherwise the leaf-wise Poincaré metric is identically zero; an example of this is provided by a linear rank one foliation on an abelian surface.

In order to prove the theorem, we first study *covering tubes*. Let $T \subset X''$ be an embedded disk transverse to $\mathcal{F}$. Then, for every $q \in T$, we call L_q the leaf of $\mathcal{F}$ through q and $\widetilde{L}_q$ its universal covering. The next proposition proves that these universal coverings glue together to a smooth complex surface U_T, called *covering tube*:

Proposition 4.3.1. *There exists a smooth complex surface U_T such that:*

1. *U_T admits a submersion $P : U_T \to T$ onto T with connected and simply connected fibres $P_t = P^{-1}(t)$ and P admits a holomorphic section $s : T \to U_T$;*

2. *U_T admits an immersion $\pi : U_T \to X'$ such that $\pi(P_t) = L_t$, $\pi(s(t)) = t$, and $\pi\big|_{P_t} : (P_t, s(t)) \to (L_t, t)$ is the universal covering of L_t with base point t.*

Sketch of the proof. The idea is to first construct a smooth complex surface V_T fibered over T, which satisfies all the desired properties except that its fibers are holonomy coverings of leaves rather than universal ones. Now, we want to prove that the fibers of V_T are simply connected, and thus we take an embedded cycle γ in a fiber H and a deformation of it γ' in a nearby fiber H' such that $\gamma' \sim 0$ in H', i.e. it bounds a disk $\Gamma' \subset H'$. We want to prove that γ bounds a disk $\Gamma \subset H$. We construct two sets

$$\Omega_\varepsilon = \varepsilon\text{-neighborhood}(\{(z,w) : z = 0, |w| \leq 1\} \cup \{(z,w) : z \in [0,1], |w| = 1\})$$

$$\hat{\Omega}_\varepsilon = \varepsilon\text{-neighborhood}(\{(z,w) : z \in [0,1], |w| \leq 1\})$$

and note that $\hat{\Omega}_\varepsilon$ is the envelope of holomorphy of Ω_ε. We construct a holomorphic embedding $i : \Omega_\varepsilon \to V_T$, for $\varepsilon > 0$ sufficiently small, that maps vertical fibers of Ω_ε into fibers of V_T, the set $\{z = 1, |w| = 1\}$ into γ, and the set $\{z = 0, |w| \leq 1\}$ into Γ'. We post-compose i with the immersion $\pi : V_T \to X' \subset X$, and since X is algebraic we can extend to a meromorphic map $\hat{j} : \hat{\Omega}_\varepsilon \dashrightarrow X$. We want to prove that $\hat{j}$ has no indeterminacy points thanks to the nefness of $K_{\mathcal{F}}$. In order to do so, we prove

Lemma 4.3.1. *Let $\mathcal{F}$ be a nef foliation on a surface X, and let*

$$f : \mathbb{D} \times \mathbb{D} \dashrightarrow X$$

be a meromorphic map such that

1. *f is an immersion outside its indeterminacy points;*

2. *$f^*(\mathcal{F})$ is the vertical foliation on $\mathbb{D} \times \mathbb{D}$.*

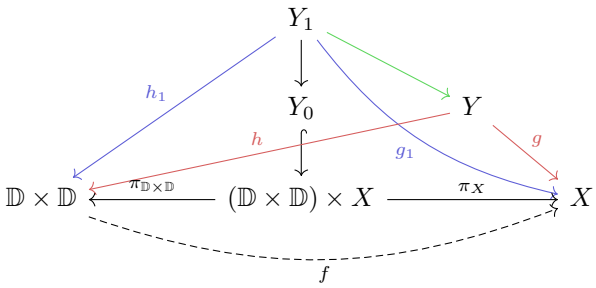
Then f is holomorphic.

By the lemma, $\hat{j}$ is holomorphic, and $\hat{j}(\{z = 1, |w| \leq 1\})$ can be lifted to V_T, giving a disk Γ with $\partial\Gamma = \gamma$. We do the same with $\gamma \subset H$ smooth cycle not necessarily embedded and prove that it is homotopic to zero; then, the fibrewise universal covering U_T of V_T, with base points on T is Hausdorff, and thus it is a complex surface U_T with all the desired properties.

□

Sketch of the proof of lemma 4.2.1. The nefness hypothesis is crucial in the proof of the lemma: indeed, suppose that f has one indeterminacy point $p = (0,0)$. Then, we can construct a normal surface Y by taking the resolution Y_1 of the

graph of f, Y_0, and then we obtain Y from Y_1 by contracting the trees of rational curves which are contracted by the projection $g_1 : Y_1 \to X$:



Thus, Y is equipped with a with a projection $g : Y \to X$ which contracts no curve, and a bimeromorphic projection $h : Y \to \mathbb{D}\times\mathbb{D}$ which contracts over $(0,0)$ a union of rational curves $R = \bigcup_{j=1}^{n} R_j$, with negative definite intersection form.

Then, we can define the foliation $\mathcal{G}$ on Y as $\mathcal{G} = g^*(\mathcal{F}) = h^*(\text{vertical foliation})$, and we observe that R is tangent to $\mathcal{G}$. We compute $K_{\mathcal{G}}$ using both g and h: since $K_{\mathcal{F}}$ is nef, $K_{\mathcal{G}}$ also is, and on the other hand the vector field generating the vertical foliation on $\mathbb{D}\times\mathbb{D}$, $\frac{\partial}{\partial w}$, can be pulled back by h to obtain a meromorphic vector field v, whose zero and polar divisor are contained in R. Hence, $K_{\mathcal{G}} = \mathcal{O}_Y(\sum_{j=1}^{n} m_j R_j)$, where $m_j \in \mathbb{Z}$ is negative if v vanishes on R_j at order $-m_j$ and is positive if v has a pole along R_j of order m_j. Since $K_{\mathcal{G}} \cdot R \geq 0$ and $(R_i \cdot R_j)_{ij}$ is negative definite, we have that $-\sum_{j=1}^{n} m_j R_j$ is effective, thus $m_j \leq 0$ for every j, and v is holomorphic; but this is a contradiction, since the local flow of v preserves R and projects on $\mathbb{D}\times\mathbb{D}$ to the local flow of $\frac{\partial}{\partial w}$, which does not fix $(0,0)$. □

Now, on each fiber of U_T we put its canonical Poincaré metric, and we have to prove that this leaf-wise Poincaré metric has positive curvature. We will frequently identify T with its image $s(T) \subset U_T$, and we shall denote with $D(z,r)$ the disk centered at z of radius r. We shall need the next result, which we will give an idea of proof later on:

Lemma 4.3.2 (Main Lemma)**.** *Let $\mathcal{F}$ be a nef foliation on X and let $P : U_T \to T$ be a covering tube associated to $\mathcal{F}$. Then, for every $z \in T$, there exists a finite subset $I_z \subset (0, d(z,\partial T))$ such that for every $r \in (0, d(z,\partial T))\backslash I_z$ the following holds: for every compact $K \subset P^{-1}(\partial D(z,r))$ there exists a real analytic 2-torus $S \subset P^{-1}(\partial D(z,r))$ such that:*

1. *S is transverse to the fibers of $P^{-1}(\partial D(z,r)) \to (\partial D(z,r))$, and it encloses K, i.e. for every $z' \in \partial D(z,r)$, the intersection $S \cap P^{-1}(z')$ is a circle which bounds on $P^{-1}(z')$ a disk containing $K \cap P^{-1}(z')$;*

2. *S is in $P^{-1}(\overline{D(z,r)})$ the boundary of a real analytic Levi-flat hypersurface M_S, disjoint from T and filled by disjoint images of holomorphic sections $s_\vartheta : \overline{D(z,r)} \to P^{-1}(\overline{D(z,r)}, \vartheta \in \mathbb{S}^1$, with boundary values on S.*

4.3.1 Proof of the main theorem

Let $P : U_T \to T$ be a covering tube associated to $\mathcal{F}$, and fix the Poincaré metric on the fibres of P, which is the pullback of the leaf-wise Poincaré metric on X thanks to property *2.* of Proposition 4.1. Since $T \subset X''$, we can take local coordinates (z, w) around $T \subset U_T$ such that $T = \{w = 0\}$, $P(z, w) = z$, and let F be the local weight of the metric in these coordinates. As already observed, F is upper semi-continuous; thus, we only need to prove that the restriction of F to T is subharmonic.

We take $z \in T$ and r as in the Main Lemma. Then, for every $S \subset P^{-1}(\partial D(z, r))$, let Ω_S be the open subset of $P^{-1}(\overline{D(z, r)})$ bounded by M_S and containing $\overline{D(z, r)} \subset U_T$. The fibres of $P : \Omega_S \to \overline{D(z, r)}$ are disks, on which we put the Poincaré metric, and we denote by F_S the corresponding local weight. Since $\partial\Omega_S = M_S$ is Levi-flat, from [20], corollary 3, we have that F_S is plurisubharmonic, and thus

$$F_S(z, 0) \leq \frac{1}{2\pi} \int_0^{2\pi} F_S(z + re^{i\theta}, 0) d\theta.$$

On the other hand, since the fibers of Ω_S are inside the fibers of U_T, we also have

$$F(z, 0) \leq F_S(z, 0).$$

We can vary the compact set K in the Main Lemma to obtain a sequence of tori $\{S_j\}$ such that for every $z' \in \partial D(z, r)$, the sequence $\{\Omega_{S_j} \cap P^{-1}(z')\}$ is increasing and exhausting in $P^{-1}(z')$. Thus, the corresponding weights F_{S_j} decreasingly converge to F on $\partial D(z, r)$. Therefore,

$$F(z, 0) \leq \lim_{j \to +\infty} F_{S_j}(z, 0) \leq \lim_{j \to +\infty} \frac{1}{2\pi} \int_0^{2\pi} F_{S_j}(z + re^{i\theta}, 0) d\theta =$$

$$= \frac{1}{2\pi} \int_0^{2\pi} F(z + re^{i\theta}, 0) d\theta,$$

which proves the subharmonicity of $F(\cdot, 0)$, since z, T, r are arbitrary, unless $F \equiv -\infty$, but in this last case all the leaves are parabolic. This completes the proof.

4.3.2 Sketch of the proof of the main lemma

In this section, we will look at the principal ideas of the proof of the main lemma.

We start by defining a real analytic Kähler metric ω on X, in the orbifold sense: around a point in $\mathrm{Sing}(X)$, ω is the cyclic quotient of a Kähler metric on the local covering $\mathbb{B}^2$. Now, we fix a covering tube $P : U_T \to T$, with immersion $\pi : U_T \to X$ mapping P-fibers onto leaves. We have a real analytic Kähler metric $\pi^*\omega$ on U_T, which we will denote also by ω.

Our first goal is to define the set I_z that appears in the statement of the main lemma. To do so, we look at the volume of the universal coverings of the leaves. Suppose that $\widetilde{L} \subset U_T$ is a fiber of P, projecting onto the leaf L;

- if L is not algebraic, then the volume of $\widetilde{L}$ is infinite, where $vol(\widetilde{L}) = \int_{\widetilde{L}} \omega$; indeed, a non-algebraic leaf accumulates on a regular point of the foliation, and thus the volume cannot be finite.
- If L is algebraic and $\pi_1(L)$ is infinite, then $\widetilde{L} \neq \mathbb{C}P^1$, and $vol(\mathbb{C}) = vol(\mathbb{D}) = +\infty$, thus $\widetilde{L}$ has infinite volume.
- Thus, the only leaves whose universal covering has a finite volume are algebraic leaves whose closure is a rational curve; moreover, this curve must contain exactly one singularity of $\mathcal{F}$ and at most one singularity of X. Indeed, concerning foliation singularities, if there were two or more then the fundamental group would be a direct sum of copies of $\mathbb{Z}$, thus infinite, while there cannot be less than one since by adjunction formula $K_{\mathcal{F}}$ would fail to be nef:

$$K_{\mathcal{F}} \cdot L = -\chi(L) + Z(\mathcal{F}, L) = -2.$$

Regarding surface singularities,

If the set of the leaves described at the third point of the list is infinite, then $\mathcal{F}$ has a rational first integral due to Jouanolou's theorem (see [3], p.75) and is a pencil of lines; in this case, we have $U_T \simeq T \times \mathbb{C}$ and everything becomes trivial. Therefore, from now on we suppose that $\mathcal{F}$ *only has a finite number of these leaves.*

Thanks to this assumption, we have that T cuts the set of these leaves at a finite set of points, so we can define

$$T_0 = \{z \in T | vol(P^{-1}(z)) = +\infty\}$$

and this set is cofinite in T, meaning $|T \backslash T_0| < +\infty$. For every $z \in T$, we define

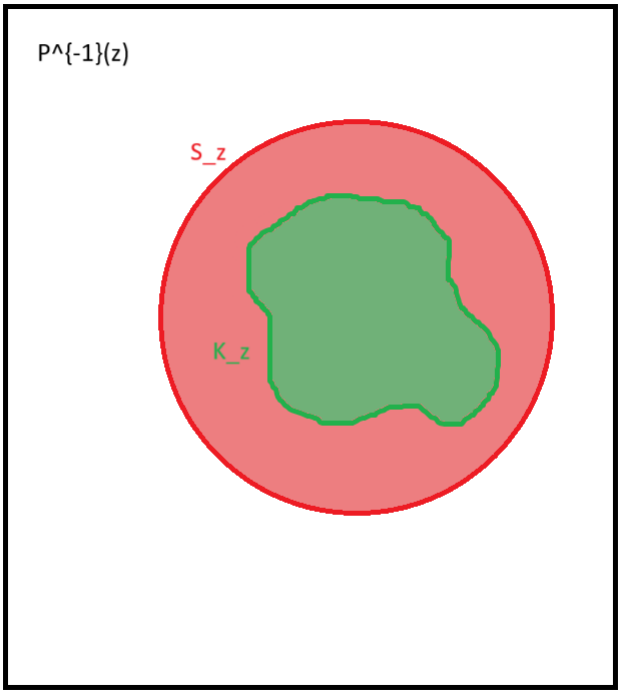
$$I_z = \{r \in (0, d(z, \partial T)) | \partial D(z, r) \not\subset T_0\};$$

thus, I_z is a finite set, with $|I_z| \leq |T \backslash T_0|$. This is precisely the set we were looking for in the first part of the statement.

We now fix $z \in T$, $r \in (0, d(z, \partial T) \backslash I_z$, so that $\partial D(z, r) \subset T_0$ and thus every point $z' \in \partial D(z, r)$ is such that $P^{-1}(z')$ has infinite volume. Since z and r are fixed, we are going to denote $D := D(z, r)$. We now take a compact set $K \subset P^{-1}(\partial D)$. Without loss of generality, we assume that K contains all the points of $T \subset U_T$ over ∂D. Take $z \in \partial D$ and a torus $S \subset P^{-1}(\partial D)$ enclosing K; we shall denote

- $K_z = K \cap P^{-1}(z)$;
- $T_z = T \cap P^{-1}(z)$;

- $S_z = S \cap P^{-1}(z)$;
- $S_z^0 =$ bounded component of $P^{-1}(z) \backslash S_z$.



The next lemma ensures the existence of a real analytic torus S as per the statement of the Main Lemma.

Lemma 4.3.3. *There exists a real analytic torus $S \subset P^{-1}(\partial D)$, transverse to the fibers of $P : P^{-1}(\partial D) \to \partial D$, enclosing K and such that the function*

$$z \mapsto vol(S_z^0), \quad z \in \partial D$$

is constant.

Our goal is now to construct a real analytic Levi-flat hypersurface $M_S \subset P^{-1}(\overline{D})$ with $\partial M_S = S$, where S is any torus provided by the previous lemma. In order to do so, we shall need the continuity method, and thus we will construct a real analytic family of tori thanks to the next lemma:

Lemma 4.3.4. *If $S \subset P^{-1}(\partial D)$ is as in Lemma 2, then there exists a real analytic family of tori $S_t \subset P^{-1}(\partial D)$, $t \in (0, 1]$, such that*

1. *every S_t is transverse to the fibers of $P : P^{-1}(\partial D) \to \partial D$ and the function $z \mapsto vol((S_t)_z^0)$ is constant;*
2. *every S_t encloses $T \cap P^{-1}(\partial D)$, and S_t degenerates to the circle $T \cap P^{-1}(\partial D)$ as $t \to 0$;*

3. $S_1 = S$.

We should note that, by "real analytic family of tori", we mean that there exists a real analytic embedding $f : \partial D \times \overline{\mathbb{D}} \to P^{-1}(\partial D)$ such that $S_t = f(\partial D \times \{|z| = t\})$ for every $t \in (0, 1]$, and $T \cap P^{-1}(D) = f(\partial D \times \{0\})$. We will later use a similar terminology, concerning Levi-flat hypersurfaces.

Definition 4.3.1. *Let $t_0 \in (0, 1]$; then, we will say that $\{S_t\}_{t\in(0,t_0]}$ has a Levi-flat extension if there exists a real analytic family of Levi-flat hypersurfaces $\{M_t\}_{t\in(0,t_0]}$ in $P^{-1}(\overline{D})$, such that:*

1. *$\partial M_t = S_t$ for every $t \in (0, t_0]$;*
2. *M_t is filled by disjoint holomorphic disks which are images of holomorphic sections of P over $\overline{D}$ with boundary values on S_t;*
3. *M_t degenerates to the disk $T \cap P^{-1}(\overline{D})$ as $t \to 0$.*

A similar definition holds for the open family $\{S_t\}_{t\in(0,t_0)}$.

Our aim is to prove that $\{S_t\}_{t\in(0,1]}$ has a Levi-flat extension; let us first see a couple of remarks.

Remark 4.3.1.

- For t_0 sufficiently small, the family $\{S_t\}_{t\in(0,t_0]}$ has a Levi-flat extension
- Since the family is real analytic, a Levi-flat extension $\{M_t\}_{t\in(0,t_0]}$ can be analytically prolonged to a family of Levi-flats $\{M_t\}_{t\in(0,t_1)}$ for some $t_1 > t_0$, and then $\{M_t\}_{t\in(0,t_1)}$ will be a Levi-flat extension of $\{S_t\}_{t\in(0,t_1)}$.

Thus, we reduce ourselves to proving the following: given $t_0 \in (0, 1]$ such that $\{S_t\}_{t\in(0,t_0)}$ has a Levi-flat extension, then also $\{S_t\}_{t\in(0,t_0]}$ has a Levi-flat extension. In other words, our aim is to construct the Levi-flat hypersurface M_{t_0}.

If $\{M_t\}$ is a Levi-flat extension of $\{S_t\}$, denote by Ω_t the open subset of $P^1(\overline{D})$ bounded by M_t and containing $T \cap P^{-1}(\overline{D})$. Thus, Ω_t is foliated in a real analytic way by holomorphic disks, images of holomorphic sections over $\overline{D}$, for which the following strong form of uniqueness holds: if $s_0 : \overline{D} \to \Omega_t$ is a section with $s_0(\partial D) \subset S_{t'}$ for some $t' < t$, and with $s_0(\overline{D}) \cap T = \emptyset$, then $s_0(\overline{D}) \subset M_{t'}$.

In order to extend the family of Levi-flat extensions up to t_0, we need an "a priori estimate" for the volume of disks filling $M_t, t < t_0$:

Lemma 4.3.5. *Let $\{M_t\}_{t\in(0,t_0)}$ be a Levi-flat extension of $\{S_t\}_{t\in(0,t_0)}$. Then, there exists a constant $C > 0$ such that*

$$vol(D_t) \leq C$$

for every holomorphic disk $D_t \subset M_t, \partial D_t \subset S_t$, and for every $t \in (0, t_0)$.

However, from this lemma we are not able to prove the convergence of the discs $D_t \subset M_t$ as $t \to t_0$, since U_T is non-compact. But we have an immersion $\pi : U_T \to X$, and thus we can define $D'_t = \pi(D_t)$ are immersed disks with boundaries on the immersed tori $S'_t = \pi(S_t)$, and $vol(D'_t) = vol(D_t) \leq C$. Thus, thanks to Bishop-Gromov compactness theorem, given any sequence $\{D_{t_n} \subset M_{t_n}\}_{n\in\mathbb{N}}$, with $t_n \to t_0$, we may extract a convergent subsequence $\{D_n\}$, which converges as $n \to +\infty$ to a "disk with bubbles" $D'_\infty \cup B_1 \cup ... \cup B_k$, where D'_∞ is a disk with boundary on S'_{t_0}, and each $B_j \subset X$ is either a rational curve or another disk with boundary on S'_{t_0}. By convergence, we mean that if $s_n : \overline{D} \to U_T$ is the section such that $s_n(\overline{D}) = \overline{D}_n$, and if $s'_n = \pi \circ s_n : \overline{D} \to X$, then there is a holomorphic map $s'_\infty : \overline{D} \to X$ and a finite set $F \subset \overline{D}$ such that $s'_\infty(\overline{D}) = \overline{D'_\infty}$ and $s'_n\big|_{\overline{D}\backslash F}$ converges to $s'_\infty\big|_{\overline{D}\backslash F}$ uniformly on compact subsets of $\overline{D}\backslash F$ as $n \to +\infty$.

Lemma 4.3.6. *There exists a section $s_\infty : \overline{D}\backslash F \to U_T$ such that $s'_\infty\big|_{\overline{D}\backslash F} = \pi \circ s_\infty$ uniformly on compact subsets of $\overline{D}\backslash F$. Moreover, $F \subset D$.*

Now, we fix a real vector field v on $P^{-1}(\partial D)$, which points inwards the tori S_t and is tangent to the fibers of P. If we restrict v to the boundary of D_n, then v gives a trivialization of $\pi^*(T_\mathcal{F})$: indeed, v is tangent to the fibers of P, which are universal coverings of leaves. By composing with π, we therefore have that v gives a trivialization of $(s'_n)^*(T_\mathcal{F})$ on D. Since the hypersurface $M_n \supset D_n$ is Levi-flat, we have that $v\big|_{\partial D_n}$ can be extended to D_n as a non-vanishing vector field tangent to the P-fibers. Hence, the degree is zero, and by dualizing we obtain $(s'_n)^*(K_\mathcal{F}) \cdot D = 0$.

We can do the same with $(s'_\infty)^*(K_\mathcal{F}) \cdot D$, but now we cannot conclude that this degree is zero since s_∞ is not defined over F.

Lemma 4.3.7.

$$(s'_\infty)^*(K_\mathcal{F}) \cdot D \geq 0,$$

with strict inequality if and only if $F \neq \emptyset$.

On the other side, by nefness of $K_\mathcal{F}$ we have

$$(s'_n)^*(K_\mathcal{F}) \cdot D - (s'_\infty)^*(K_\mathcal{F}) \cdot D = K_\mathcal{F} \cdot B \geq 0,$$

thus $(s'_\infty)^*(K_\mathcal{F}) \cdot D = 0$ and $F = \emptyset$, so s_∞ is defined over the full $\overline{D}$.

In other words, we just proved that, given any sequence of disks $\{\overline{D_{t_n}} \subset M_{t_n}\}$, with $\partial D_{t_n} = S_{t_n}$, as $t_n \to t_0$ we may construct a convergent subsequence $\overline{D_n}$, whose limit $\overline{D_\infty}$ is the image of a section $s_\infty : \overline{D} \to U_T$, holomorphic up to the boundary. Thus, if we can glue together these limits to a real analytic Levi-flat hypersurface M_{t_0} with $\partial M_{t_0} = S_{t_0}$ and such that $\{M_t\}_{t\in(0,t_0]}$ is a Levi-flat extension of $\{S_t\}_{t\in(0,t_0]}$, we are done.

In order to do so, we refer to [2], chapter 5: according to this paper, if $D_\infty \subset U_T$ is a limit disk with border on S_{t_0}, then D_∞ belongs to a unique real analytic family of disks D^ε_∞, where all the disks have borders on S_{t_0}, such that $D^0_\infty = D_\infty, \varepsilon \in \mathbb{R}, |\varepsilon| << 1$; this family has winding number zero, thus it

can be deformed to a family $D_n^\varepsilon \subset U_T$ with borders on S_{t_n}, $D_n^0 = D_n \subset M_{t_n}$, which must coincide with the family already provided by M_{t_n} by the uniqueness property we discussed before. Hence, the disks D_∞^ε are contained in the closure of $\bigcup_{t\in(0,t_0)} M_t$ and they also are limits of disks inside the lower Levi-flats, and thus M_t converges, as $t \to t_0$, to the real analytic Levi-flat M_{t_0}, with border on S_{t_0} and all the desired properties. We have now proved the Main Lemma.

4.4 Regularity results

In this section, we will briefly present a result concerning the regularity of the leaf-wise Poincaré metric that is independent from the Main Theorem, but will be useful in proving the conjecture. Since there exists a hyperbolic leaf, we have that the parabolic leaves are all algebraic: indeed, if we had a non-algebraic parabolic leaf, then there would exist a Zariski-dense entire curve $f : \mathbb{C} \to X$, tangent to $\mathcal{F}$, which is a transcendental object, not contained in any algebraic curve. From McQuillan's tautological inequality ([15], fact V.4.2), and from the nefness of $K_\mathcal{F}$, we have that $K_\mathcal{F}$ has to be numerically trivial along the entire curve; but this cannot happen due to the fact that $K_\mathcal{F}$ has positive curvature on the hyperbolic leaf, and so any transcendental entire curve tangent to $\mathcal{F}$ contradicts the tautological inequality. Thus, the regularity problem makes sense only if the closure of parabolic leaves, which we denote by $\mathcal{P} \subset X$, is a (possibly empty) algebraic curve. Moreover, if the foliation is a fibration everything becomes trivial, so we may restrict ourselves to the case where there is a finite number of algebraic leaves; equivalently, we say that the closure of algebraic leaves, $\mathcal{A} \subset X$, is a possibly empty algebraic curve.

Proposition 4.4.1. *Let $\mathcal{F}$ be a nef foliation on X, and suppose that $\mathcal{A}$ is an algebraic curve containing $\mathcal{P}$. Then, the leaf-wise Poincaré metric is continuous on $X\backslash(\mathcal{A} \cup \mathrm{Sing}(\mathcal{F}))$.*

4.5 Answer to the question

In this section, we are finally ready to negatively answer the Question we posed at the end of Section 3:

Question

Is there any non-abundant foliation that is not a Hilbert modular foliation?

In other words, we want to prove the following

Theorem 4.5.1. *Let $\mathcal{F}$ be a nef foliation with reduced singularities on a compact complex algebraic surface X, and suppose* $\mathrm{kod}(\mathcal{F}) = -\infty$. *Then, $\mathcal{F}$ is a Hilbert Modular foliation.*

Let $\mathcal{F}$ satisfy the hypothesis of the theorem. At the end of Section 3 we showed that $H^1(X, \mathcal{O}_X) = 0$; moreover, the first Chern class of the canonical

bundle, $c_1(K_{\mathcal{F}}) \in H^2(X, \mathbb{R})$, is not trivial: indeed, if that were not the case then $K_{\mathcal{F}} \in \ker(c_1) = Pic^0(X)$, and the latter is trivial since it has dimension $h^1(X, \mathcal{O}_X)$. Moreover, $c_1(K_{\mathcal{F}})^2 = 0$ since $K_{\mathcal{F}}$ is nef and not big (otherwise, $\mathrm{kod}(\mathcal{F}) = 2$). In addition to this, $\mathcal{F}$ has a finite number of algebraic leaves: indeed, if this were not the case, then by a corollary of Jouanolou's theorem ([3], Corollary 6.1) $\mathcal{F}$ would be a fibration, but from the previous classification results we reach a contradiction with $\mathrm{kod}(\mathcal{F}) = -\infty, \nu(\mathcal{F}) = 1$. Finally, $\mathcal{F}$ has a finite number of parabolic leaves, and they are all algebraic: see [15], Corollary V.1.4.

As before, the idea for the proof of the theorem is to look for the companion foliation $\mathcal{G}$ of $\mathcal{F}$ in the Kernel of a closed positive current Ω, which is the curvature of a singular hermitian metric on $K_{\mathcal{F}}$; we know that, if we can find this foliation $\mathcal{G}$ and it is holomorphic, then we have proved the theorem. The goal of this section is thus to prove that, if we take the canonical Poincaré metric as singular hermitian metric on $K_{\mathcal{F}}$, then the foliation $\mathcal{G}$ is well-defined and holomorphic. Note that we can take the canonical Poincaré metric because, in the Main Theorem, we proved that it has positive curvature.

In order to continue, we need a regularity result:

Proposition 4.5.1. *Let $\mathcal{F}$ be a nef foliation with reduced singularities and Kodaira dimension $-\infty$. Let Ω be the curvature of the canonical Poincaré metric. Then,*

1. *Ω has everywhere vanishing Lelong number;*
2. *Ω is absolutely continuous (i.e. it is a $(1,1)$-form with L^1_{loc} coefficients);*
3. *$\Omega \wedge \Omega \equiv 0$ in a punctual sense.*

Sketch of the proof. The first step of the proof is to consider the Siu decomposition of Ω ([7], 2.18):

$$\Omega = \sum \lambda_j \delta_{C_j} + \Omega_{res} = \Omega_{alg} + \Omega_{res},$$

with $\lambda_j > 0, \delta_{C_j}$ = integration current over the algebraic curve $C_j \subset X$, and Ω_{res} = closed positive current with vanishing Lelong number outside a countable set. We can then prove that $\Omega_{alg} = 0$ and thus $\Omega = \Omega_{res}$, and from this and the fact that $[\Omega]^2 = 0$ we have that the Lelong number of Ω is zero everywhere, and not just outside a countable subset ([7], 2.16).

The proof of the second thesis is the part where the fact that Ω is the curvature of the canonical Poincaré metric comes into play. From the fact that Ω has everywhere vanishing Lelong numbers, we have ([7], 9.1) that there exists a sequence of smooth closed $(1,1)$-forms Ω_k on X, $k \in \mathbb{N}$, such that

1. $[\Omega_k] = [\Omega]$ for every k;
2. $\Omega_k \geq -\frac{1}{k}\omega$ for every k, where ω is a fixed Kähler form on X;
3. $\Omega_k \xrightarrow{weak} \Omega$ as $k \to +\infty$;

4. $\Omega_k \wedge \Omega \xrightarrow{weak} 0$ as $k \to +\infty$.

If we now take a local chart $U \subset X''$, with coordinates (z, w) such that $\mathcal{F} = \{dz = 0\}$, then we have a plurisubharmonic function F on U such that

$$\Omega = \frac{i}{2\pi}\partial\overline{\partial}F \quad \text{and} \quad F_{w\overline{w}} = e^F,$$

and moreover F is continuous by Proposition 4.3.1. By construction ([7], chapter 9), the regularizing forms Ω_k can be chosen in such a way that, on U, we have $\Omega_k = \frac{i}{2\pi}\partial\overline{\partial}G_k$ and $G_k \searrow F$. Thus, for every $\varphi \in C_c^\infty(U)$, and by 4. above, we have

$$\int_U F \cdot (i\partial\overline{\partial}\varphi) \wedge \Omega = \lim_{k\to+\infty} 2\pi \int_U \varphi \cdot \Omega_k \wedge \Omega = 0.$$

After this, we take the Lebesgue decomposition of Ω into absolutely continuous and singular parts, $\Omega = \Omega_{ac} + \Omega_{sg}$, and we prove that $\Omega_{sg}\big|_U \equiv 0$; thus, we have proved that $Supp(\Omega_{sg})$ is contained in the union of algebraic leaves and the singularities of the foliation; from the vanishing of the Lelong numbers of Ω, it follows that the Lelong numbers of Ω_{sg} are well defined and everywhere vanishing, therefore from Lebesgue's density theorem we deduce that Ω_{sg} is identically zero, and thus that Ω is absolutely continuous.

Finally, the vanishing of the pointwise product $\Omega \wedge \Omega$ follows from $[\Omega]^2 = 0$ and [9], 9.5. □

Proposition 4.5.2. *Let F be a locally bounded plurisubharmonic function on $\mathbb{D} \times \mathbb{D}$ such that*

1. *$\Omega = \frac{i}{2\pi}\partial\overline{\partial}F$ is absolutely continuous;*
2. *$\Omega \wedge \Omega \equiv 0$;*
3. *$F_{w\overline{w}} = e^F$.*

Then, there exists a holomorphic foliation $\mathcal{G}$ on $\mathbb{D} \times \mathbb{D}$ in the kernel of Ω and transverse to the verticals.

Corollary 4.5.1. *Let $\mathcal{F}$ be a nef foliation with reduced singularities, and with Kodaira dimension $-\infty$. Then, $\mathcal{F}$ is a Hilbert modular foliation.*

Proof. By propositions 4.3.1, 4.4.1, 4.4.2 we have that a holomorphic foliation $\mathcal{G}$ in the kernel of Ω and transverse to $\mathcal{F}$ certainly exists, outside of algebraic leaves and the singularities of $\mathcal{F}$. To prove Theorem 4.4.1, we only need to check that this foliation $\mathcal{G}$ extends to the full X; we aim to extend it to X'', whose complement in X has codimension 2.

Let $U \subset X''$ be a local chart, with local coordinates (z, w) such that $\mathcal{F} = \{dz = 0\}$. Let F be the local weight of the canonical Poincaré metric, so that the kernel of $\Omega = \frac{i}{2\pi}\partial\overline{\partial}F$ is expressed by

$$\frac{dw}{dz} = -F_{z\overline{w}}e^{-F} = -G.$$

Since F has zero Lelong numbers, we have $e^{-F} \in L^2_{loc}$, and the same holds for $F_{z\overline{w}}$ for $|F_{z\overline{w}}|^2 = F_{z\overline{z}}e^F \in L^1_{loc}$, hence $G \in L^1_{loc}$. Moreover, we already know that G is holomorphic outside of $\Gamma = U \cap \{\text{algebraic leaves}\}$, which is a finite union of verticals in U. We obtain that G is meromorphic on U, with at most first order poles along Γ, which means that $\mathcal{G}$ holomorphically extends to the whole U. $\square$

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